Lensing of moiré patterns by atomic area defects: topological node charge, dressed charge, and the ring-traction identity

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A point defect in one layer of a bilayer—a vacancy cluster, an inserted island, or a rotated patch—bends the moiré domain-wall network far beyond its own size and can remove moiré nodes. We ask, within a continuum model of the relaxed registry, how many nodes a defect removes and where they go. AA nodes are the degree density of the registry map $u : \Sigma \to T^2$ and obey a Gauss law in which every defect core carries an integer charge. An area or torque defect with complex charge C deforms the moiré through the lens map $\beta = z - m/\bar{z}$ with mass $m = C/\kappa$, where $\kappa = \theta e^{i\alpha}$ encodes twist or mismatch; for real m this is the point-mass gravitational lens. Averaged over the position of the defect in the moiré cell, the removed node charge $\theta n_{\rm miss}$ splits into a far share $\propto \cos\varphi_m$ and a core-bound share $\propto 1 - \cos\varphi_m$, with $\varphi_m = \arg C - \alpha$. Relaxed simulations over $T = \theta n_{\rm miss} = 0.5$–$6$ bind the predicted integers when relaxed from the lens field. The far charge is set by a dressed charge $C_{\rm eff} = \sum C + C_{\rm reg}$, the force dipole of the reorganized registry forces. For a single defect this dressing screens the lens mass, $m_{\rm eff} = \rho m$: the moiré near the core over-twists and keeps part of the removed nodes, and ρ falls from 1 (no registry) to 0.1–0.2 for strong relaxation, independent of $|C|$. A local-density theory based on the energy of sheared moirés explains the trends. For Einstein radii beyond the moiré period the lens resolves into arcs and, for $\varphi_m = 0$, a ring of uniform stacking; relaxation turns these into spiral or concentric wall textures.

I. INTRODUCTION

Two lattices stacked with a small twist angle or a small lattice mismatch form a moiré superlattice whose period exceeds the atomic spacing by a factor $1/\theta$. Since the discovery of correlated and superconducting phases in magic-angle graphene [1–3], moiré systems have become a platform for engineering band structures and interactions [4, 5]. The relevant length scale is set by the *relaxed* moiré pattern: interlayer adhesion turns the smooth interference pattern into AB/BA domains separated by shear solitons that meet at AA nodes [6–13]. Because the moiré magnifies the relative displacement of the two layers by $1/\theta$, it is extremely sensitive to strain. Twist-angle disorder, heterostrain, and local strain fields are visible directly in scanning-probe and diffraction maps [14–19].

Point defects are an equally natural source of local strain. Vacancy clusters, ion-irradiation craters, adatom patches and grain-boundary fragments are common in two-dimensional crystals [20]. A cluster of a few tens of missing atoms produces a long-ranged displacement field of order $|C|/r$. In the moiré pattern this field is amplified by $1/\theta$ into a distortion comparable to the moiré period. Two questions then arise: *how many moiré nodes does an atomic defect remove, and where does that missing node charge reside?* The answer is not obvious. The number of missing atoms is an integer, the number of moiré nodes is an integer, and yet the conversion factor between them is the continuous parameter θ.

Related questions have been studied for extended defects and smooth strain. Atomic dislocations in bilayer graphene appear as dislocations of the AB/BA domain pattern [7, 31]. Long-wavelength deformations of the moiré have been described by moiré phonons and phasons [32, 33], and uniform heterostrain reshapes the pattern and the bands [15, 16, 34]. Point defects with a monopole area or torque charge have, to our knowledge, not been treated from the viewpoint of the topology of the registry map.

In this paper we answer these questions within a minimal continuum model. Four results stand out.

(i) *Moiré nodes are topological charges.* The moiré pattern is a map u from the sample to the registry torus T^2. AA nodes are the points where u takes the AA value, counted with the sign of $\det \nabla u$; they are the degree density of u. A Gauss law fixes the node count in any region from boundary data alone, and every hole carries an integer charge.

(ii) *An atomic area defect acts as a lens for the moiré.* The far field $-C/\bar{z}$ of a defect with complex charge C (area deficit plus torque) maps the undeformed moiré coordinate β to the physical coordinate z through $\beta = z - m/\bar{z}$ with lens mass $m = C/\kappa$. For real m this is the point-mass gravitational lens equation [21, 22]; a complex m adds a rotational deflection without gravitational counterpart. Undeformed moiré nodes play the role of light sources. Sources inside a disk of radius $b_s = 2R|\sin(\varphi_m/2)|$ have no images and are swallowed by the defect. As a result, the removed node charge $2\pi\theta|C|/A_c$ splits, on average over the defect position, into a far share $\propto \cos\varphi_m$ and a local share $\propto 1 - \cos\varphi_m$, both fixed by the single phase $\varphi_m = \arg C - \alpha$.

(iii) *The far charge is dressed by the moiré.* The registry forces redistribute when the defect reorganizes the wall network, and their force dipole adds an induced charge $C_{\rm reg}$. The far node charge is fixed by

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$C_{\rm eff} = \sum C + C_{\rm reg}$, not by the atomic charge alone. $C_{\rm reg}$ depends on the integer branch selected by the pattern. For a single defect it *screens* the lens: the relaxed moiré around the core is over-twisted and retains part of the removed nodes, so the far lens mass is $m_{\rm eff} = \rho m$ with $\rho < 1$ set by the relaxation strength and the bulk stiffness, but not by $|C|$.

(iv) *Local sources do not couple through the moiré elastic constants.* We measure the elastic constants of the moiré crystal. The twisted moiré has a negative bulk modulus $K_M < 0$, the mismatch moiré a negative rotational stiffness $\chi_M < 0$. One might expect the response to a local moiré-scale source to follow from these constants. An exact ring-traction identity shows instead that such a source is driven only by the *atomic* traction on its boundary, not by the moiré pre-stress. The far node charge is then accounted for exactly by integer bookkeeping and the force-dipole dressing, although $C_{\rm reg}$ of interacting pairs is not yet predictable.

Figure 1 shows which atomic defects act as lenses, and Fig. 2 summarizes the central picture. Defects that change the transmitted area or rotation of the layer—vacancy clusters, inserted islands, rotated patches—carry a monopole charge C and lens the moiré. Area-neutral combinations produce only short-ranged distortions, and dislocations are a topologically different object: they create a moiré dislocation rather than a lens.

Section II introduces the model, Sec. III the topology of the registry map, and Sec. IV the lens map and the partition theorem. Section V treats several defects and the conservation of far charge. Section VI introduces the dressed charge, and Sec. VII shows that for a single defect it screens the lens mass. Section VIII discusses moiré elasticity and the ring-traction identity. Section IX treats lenses larger than the moiré period, and Sec. X discusses experimental consequences.

II. MODEL

The dynamical variable is the interlayer relative displacement $u(\boldsymbol{r}) \in \mathbb{R}^2$; we use complex notation $z = x+iy$, $u = u_x + iu_y$. The energy is

$$E[u] = \int_\Sigma \Big[\mu\, \varepsilon_{ij}\varepsilon_{ij} + \frac{\lambda}{2}\varepsilon_{kk}^2 + V(u)\Big] \mathrm{d}^2 r, \qquad (1)$$

with $\varepsilon = \frac{1}{2}(\nabla u + \nabla u^T)$ the strain of the total relative displacement. The background $u_b = c_{\rm AA} + \kappa z$ with $\kappa = \theta e^{i\alpha}$ describes a twist ($\alpha = 90°$) or an isotropic lattice mismatch ($\alpha = 0$); we write $\delta u = u - u_b$. In this reference the twisted background (a rotation) is stress-free, while the mismatch background carries a uniform dilational prestress. Appendix A shows when the choice of reference matters. We take $u = u_{\rm top} - u_{\rm bottom}$ and place all defects in the top layer. A defect in the bottom layer enters with $C \to -C$, i.e., $\varphi_m \to \varphi_m + 180°$. For the mismatch, $\kappa = \theta > 0$ means that the top layer has the larger lattice constant.

The registry potential is

$$V(u) = V_0 \sum_{\boldsymbol{G}} \big(1 - \cos \boldsymbol{G}\cdot u\big) + V_3 \sum_{\boldsymbol{G}} \sin \boldsymbol{G}\cdot u, \qquad (2)$$

where $\boldsymbol{G}$ runs over three reciprocal vectors of the triangular lattice Λ (lattice constant $a = 1$, cell area $A_c = \sqrt{3}/2$) and $V_3 = \sqrt{3}V_0$. This honeycomb-type registry has a maximum at $c_{\rm AA} = (a_1 + a_2)/3$ (AA stacking) and degenerate minima at AB and BA. It is C_6 symmetric about AA, including layer exchange $u \to -u$, and has a mirror symmetry [6, 8].

Iris defects. A defect at z_k is a disk of radius r_h removed from the energy. A penalty fixes the mean normal and tangential displacement of its rim, $\langle \delta u \cdot n\rangle = -\mathrm{Re}\, C/r_h$ and $\langle \delta u \cdot t\rangle = -\mathrm{Im}\, C/r_h$, while the rim shape is free. The complex charge C combines an area deficit ($C > 0$, a "closed iris"), an inserted area ($C < 0$), and a rim rotation or point torque ($\mathrm{Im}\, C \neq 0$). In an infinite medium the equilibrium field is

$$\delta u = -\frac{C}{\bar z - \bar z_k}. \qquad (3)$$

It is the gradient of a harmonic potential, so $\mathrm{div}\, \delta u = \mathrm{curl}\, \delta u = 0$ and it solves Eq. (1) for any λ, μ (pure shear). A defect that removes $n_{\rm miss}$ unit cells and is fully closed by the surrounding layer has $C = n_{\rm miss} A_c/2\pi$, as measured by the displacement flux far away. With $|C| = 4$ and $r_h = 2.4$ the core strain is of order one, so the integers bound to a core depend on the nonlinear core and are model specific.

Numerics. We minimize Eq. (1) with L-BFGS [29] on a square grid of spacing $h = 0.45$ with $u = u_b$ on the outer boundary. The standard box has half-width 72; the large boxes have half-width 148.5 (grid 661^2, about 105 nodes). Unless noted, $\theta = 1/30$, $V_0 = 0.005$, $\mu = 1$, $\lambda = -0.5$ (2D bulk modulus $K = \lambda + \mu$, $K/\mu = 0.5$), $|C| = 4$, $r_h = 2.4$. The soft bulk modulus is a deliberate choice that separates the shear and dilation sectors; the dressed-charge calculations of Sec. VI were repeated at $K/\mu = 2$. A tanh fit of the relative displacement across a wall gives the width $w \approx 0.21\sqrt{K_w/V_0}$, with $K_w = \mu$ for shear walls and $\lambda+2\mu$ for tensile walls; for our parameters $w \approx 3.0$ (twist) and 3.7 (mismatch), i.e., 0.10–0.12L_M: the moiré is strongly relaxed. The lens (Einstein) radius is $R = \sqrt{|C|/\theta} \approx 11$ and the relaxed moiré period is ≈ 31. All parameters are collected in Table VI (Appendix D).

III. TOPOLOGY OF THE REGISTRY MAP

A. Degree and the wall network

An affine registry $u = c + Az$ ($\det A \neq 0$) maps the moiré lattice $\Lambda_M = A^{-1}\Lambda$ onto Λ. It therefore induces a homeomorphism $\mathbb{R}^2/\Lambda_M \to T^2$ of degree $\mathrm{sign}\det A$; in complex notation $u = \kappa z + \eta \bar z$ gives $\det A = |\kappa|^2 - |\eta|^2$.

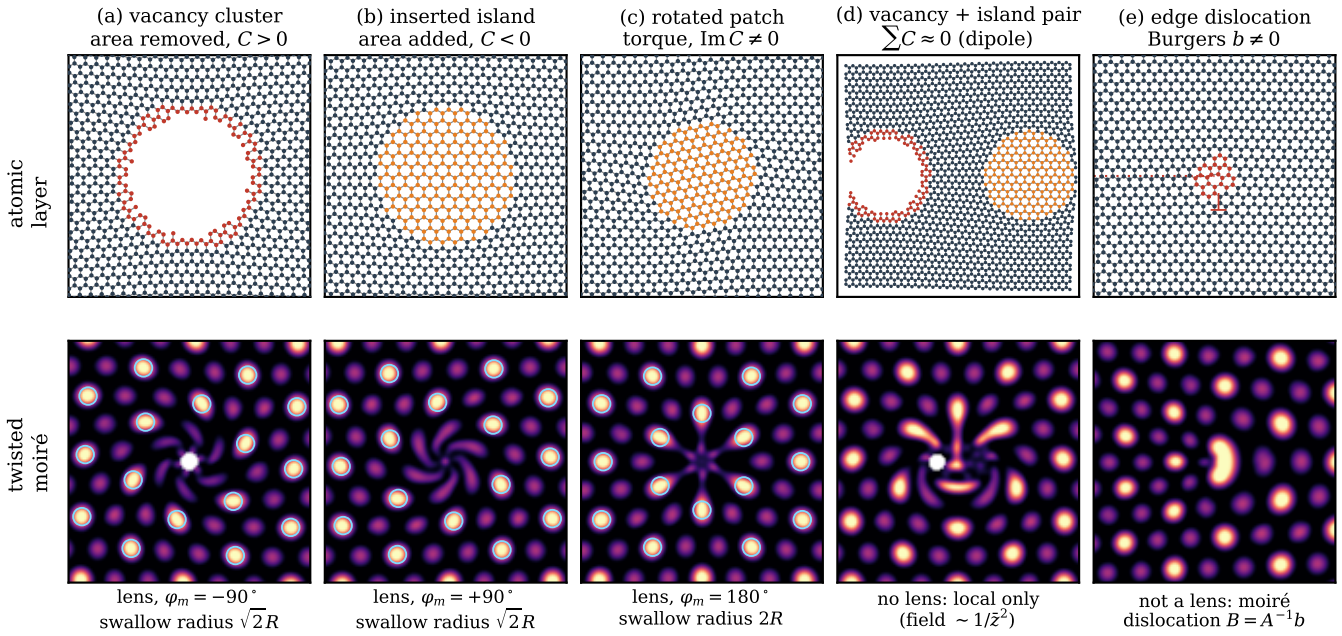


FIG. 1. Which atomic defects lens a moiré pattern. Top: one graphene layer containing (a) a vacancy cluster whose edge has moved inward so that 30% of the removed area is closed ($C > 0$), (b) an inserted island ($C < 0$), drawn as a coherent inclusion that is uniformly expanded and joins the host lattice continuously, (c) a rotated patch ($\mathrm{Im}\, C \neq 0$), (d) a vacancy cluster and an island with $\sum C \approx 0$, and (e) an edge dislocation (Burgers vector b; dotted line: cut). Atoms are displaced by the continuum displacement field with far tail $-C/\bar{z}$ (an illustration based on linear elasticity; core reconstruction is not modeled); highlighted atoms belong to the defect. Bottom: moiré formed with a second, perfect layer twisted by $\theta = 1/30$, computed directly from the atomic positions (bright: atoms on top of atoms, i.e., AA stacking; no interlayer relaxation). Circles mark the node positions predicted by the lens map, Eq. (8). Defects (a)–(c) are lenses with Einstein radii $R \approx 11$–14; (d) distorts the pattern only locally; (e) inserts a moiré dislocation $B = A^{-1}b$.

Twist and isotropic mismatch have degree $+1$; a strain dominated by $\bar{z}$ has degree -1.

Cutting T^2 into AB and BA regions gives a minimal cell decomposition (a Δ-complex): one vertex (AA), three edges, and two faces, with Euler number zero [Fig. 2(a)]. In the universal cover $\mathbb{R}^2$ the one-skeleton is the arrangement $\mathcal{L}$ of three families of lines through the AA lattice points. We define the domain walls as $u^{-1}(\mathcal{L})$ and the nodes as $u^{-1}(\mathrm{AA}+\Lambda)$. Two consequences follow:

Even valence. The lift of a closed loop with zero Burgers class is a closed curve in $\mathbb{R}^2$, so by the Jordan curve theorem it crosses each line of $\mathcal{L}$ an even number of times. Hence the number z of walls meeting at a vertex, or crossed by a loop around a defect core, is even. Ordinary AA nodes have $z = 6$; the eight- and ten-wall vertices seen near defects obey the same rule. Where all faces are triangles, Euler's theorem gives $\sum_v (6 - z_v) = 6\chi$, so a core behaves like a vertex carrying disclination charge $6 - z_{\mathrm{core}}$ [23, 24].

B. Node charge and Gauss law

A loop γ with vanishing Burgers class $\oint_\gamma \mathrm{d}u \in \Lambda$ lifts to $\mathbb{R}^2$; iris defects have zero Burgers class. We define the node charge

$$Q(\gamma) = \sum_{n \in \Lambda} \mathrm{wind}\big(\tilde{u} \circ \gamma,\ c_{\mathrm{AA}} + n\big) \in \mathbb{Z}, \tag{4}$$

which is invariant under deformations that do not cross a node. At a regular point the index of a node is $\mathrm{sign} \det \nabla u = \pm 1$: ordinary nodes N_+ and inverted nodes N_-. For a region D containing holes H_k (defect cores) on whose complement u is continuous,

$$N_+(D) - N_-(D) + \sum_k Q(\partial H_k) = Q(\partial D), \tag{5}$$

with node density $\rho_N = \det \nabla u / A_c$. The proof uses additivity of winding numbers: interior edges of a fine subdivision cancel, and a cell containing one node contributes its local degree. Because ρ_N is a null Lagrangian, $\partial_t \rho_N + \nabla \cdot j = 0$ holds identically in any continuous dynamics. Nodes therefore enter or leave only through

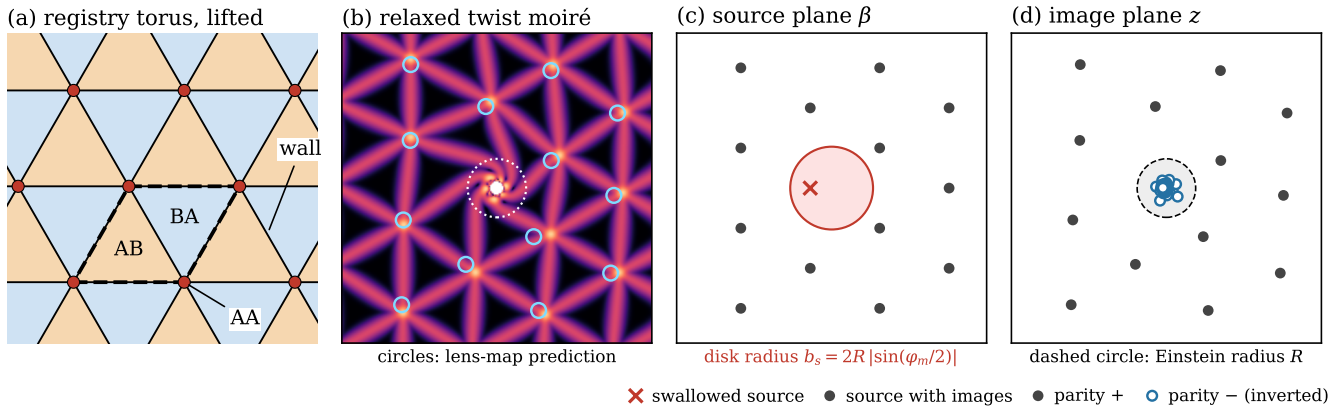


FIG. 2. Moiré nodes as lensed images. (a) Universal cover of the registry torus $T^2 = \mathbb{R}^2/\Lambda$. The three line families $\mathcal{L}$ are the domain walls, the up (down) triangles are AB (BA) stacking, and the lattice points are AA. The dashed parallelogram is one fundamental cell. (b) Relaxed twisted moiré ($\theta = 1/30$) around an area-removing defect ($C = 4$, white disk); color shows $\log_{10} V(u)$ (domain walls bright, AA nodes brightest). Circles mark the positive-parity images predicted by the lens map; the dotted circle is the Einstein radius $R = \sqrt{|C|/\theta}$. (c) Source plane β: undeformed node positions. Sources inside the disk of radius b_s (crosses) have no images. (d) Image plane z for the same lens ($\varphi_m = -90°$). Each surviving source has an ordinary image outside R (filled) and an inverted image inside R (open).

holes, or are created and annihilated in ± pairs [25]. This is the sense in which moiré nodes are conserved charges.

Valence–winding bound. If the AA points enclosed by the lift of γ have windings 0 or ±1 of one sign with total w, then

$$z(\gamma) \geq 2L_{\min}(|w|), \qquad z(\gamma) \in 2\mathbb{Z}, \tag{6}$$

where $L_{\min}(n)$ is the minimum number of lattice lines of the three families needed to cover n triangular-lattice points ($L_{\min} = 3, 5, 6, 7, 8, 9, 9$ for $n = 1, \ldots, 7$). By a Jordan-curve argument, each line through an enclosed point is crossed at least twice. The bound is $2L_{\min}$ rather than $6|w|$ because neighboring points share lines.

Counting. The discrete count used throughout (Appendix C) satisfies Eq. (5) exactly on the grid, so the zero deviation over 84 relaxed states [Fig. 3(a)] is a consistency check of the counting rather than an independent test. A single area-removing defect in the twisted moiré has $Q_{\text{hole}} = 5$ and $N_- = 4$, so it binds $q \equiv Q_{\text{hole}} - N_- = +1$ node; in the mismatch moiré $Q_{\text{hole}} = N_- = 2$, so $q = 0$ [Fig. 3(b)]. Among 444 circles around cores, 162 saturate the bound of Eq. (6) [Fig. 3(c)]: walls around cores are close to minimal.

IV. AREA DEFECTS AS MOIRÉ LENSES

A. Complex charge

For any loop around a set of defects, Eq. (3) gives

$$\oint_\gamma \delta u \cdot (n + i\,t)\,\mathrm{d}s = -2\pi \sum_k C_k. \tag{7}$$

Thus C is a conserved charge, and C and $-C$ can be created or annihilated in pairs. However, the field (3) is traceless and curl-free, so it does no first-order work on a dilatation or torque center. Two such charges therefore have no long-range elastic interaction regardless of their signs, as for Eshelby inclusions in an infinite medium [26]. Elasticity alone is sign-blind.

B. Lens map

Writing $u = c + \kappa\beta$ defines the undeformed moiré coordinate β. For one defect at the origin,

$$\beta(z) = z - \frac{m}{\bar{z}}, \qquad m = \frac{C}{\kappa} = R^2 e^{i\varphi_m}, \tag{8}$$

with Einstein radius $R = \sqrt{|C|/\theta}$ and mass phase $\varphi_m = \arg C - \alpha$. For real m this is the lens equation of a point mass, with source position β and image position z [21, 22]; $\varphi_m = 180°$ corresponds to a negative mass. For complex m the deflection $m/\bar{z}$ also rotates the images, which has no gravitational counterpart. We nevertheless keep the lens language, since the image structure is the same. The lens mass m is the defect charge transferred to the moiré level, and the undeformed moiré nodes form a perfect lattice of density $n_M = \theta^2/A_c$ in the source plane. The Jacobian $J = 1 - R^4/|z|^4$ is negative inside the Einstein circle, where images are inverted.

Image structure. Let $z = re^{i\phi}$. Then $|\beta|^2 = r^2 + R^4/r^2 - 2R^2\cos\varphi_m$. A source at $|\beta| = b$ therefore has no image if

$$b < b_s = 2R\,\big|\sin(\varphi_m/2)\big|, \tag{9}$$

and otherwise has one ordinary image at $r_+ > R$ and one inverted image at $r_- = R^2/r_+$. The disk $b < b_s$ is

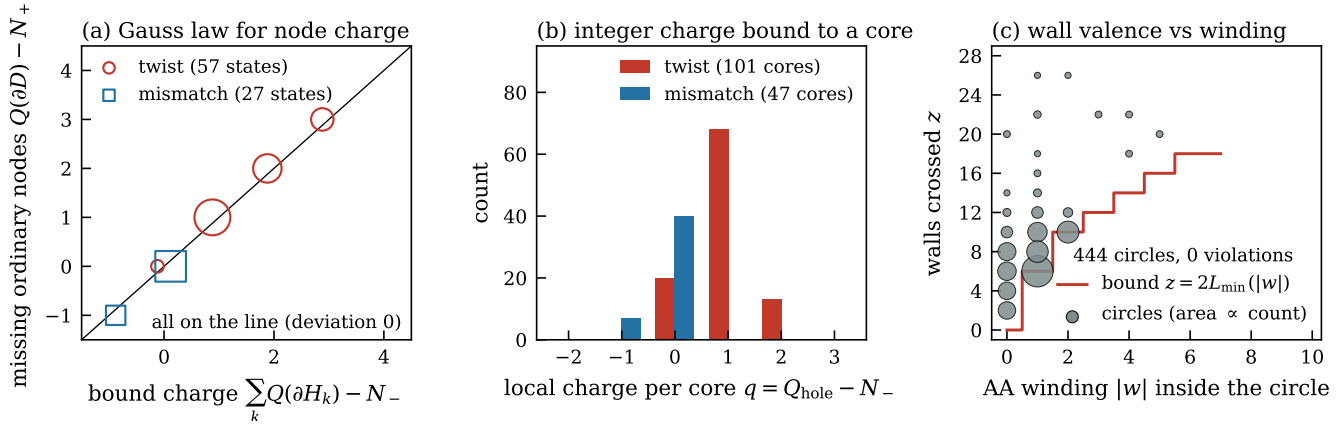


FIG. 3. Topological charges of the registry map. (a) Gauss law, Eq. (5): the ordinary-node deficit equals the charge bound to holes and inverted nodes in all 84 states. Symbol area is proportional to the number of states. (b) Distribution of the integer charge $q = Q_{\rm hole} - N_-$ bound to a single core, over all single-defect and pair states ($K/\mu = 0.5$). (c) Number of walls z (not to be confused with the coordinate z) crossed by circles of radius 5, 7, 9 around cores versus the AA winding $|w|$ they enclose. The red staircase is the bound of Eq. (6).

a shadow of the lens; it exists only for $\varphi_m \neq 0$. Its size depends on the relative phase of the defect charge and the moiré background. For an area defect in a twisted moiré, $\varphi_m = \mp 90°$ and $b_s = \sqrt{2}R$. For an area-removing defect in the larger-lattice layer of a mismatch moiré, $\varphi_m = 0$ and nothing is swallowed; in the smaller-lattice layer, $\varphi_m = 180°$ and $b_s = 2R$. Figure 2(b) shows that the predicted ordinary images coincide with the relaxed AA nodes outside the core region.

C. Partition theorem

Averaging over random translations of the moiré lattice, the expected number of sources in a region S of the source plane is $n_M|S|$. Positive images inside $|z| < \rho$ correspond one-to-one to sources with $b_s < b < b(\rho) = |\rho - m/\rho|$, and the loop $|z| = \rho$ maps onto a curve enclosing area $\pi b(\rho)^2$. For $\rho > R$ this yields

$$\Delta N_+(\rho) = T - \frac{\pi n_M R^4}{\rho^2}, \qquad T \equiv \frac{2\pi\theta|C|}{A_c},$$
$$\Delta Q(\rho) = -\frac{2\pi}{A_c}\mathrm{Re}(\bar\kappa C) + \frac{\pi n_M R^4}{\rho^2},$$
$$q = Q_{\rm hole} - N_- = T\,(1 - \cos\varphi_m), \qquad (10)$$

which are tied together by the Gauss law $q = \Delta N_+ + \Delta Q$; here we used $n_M R^2 \cos\varphi_m = \mathrm{Re}(\bar\kappa C)/A_c$. The ordinary-node deficit at infinity, $T = \theta\, n_{\rm miss}$, is independent of φ_m: *a defect missing $n_{\rm miss}$ unit cells removes $\theta n_{\rm miss}$ moiré nodes.* Of these, the share $T(1 - \cos\varphi_m)$ is swallowed and bound to the core, and the share $T\cos\varphi_m$ is spread to infinity as boundary charge ΔQ. For our parameters $T = 0.967$.

In the relaxed moiré the local share must be an integer, and Eq. (10) becomes a rounding rule. Figure 5 and Table I compare the eight defects $C = \pm 4, \pm 4i$ in both backgrounds. In seven of the eight, the bound integer q equals the rounded prediction. The exception, a torque along the twist sense ($\varphi_m = 180°$), binds $q = 1$ instead of 2.

Because $T = 0.97$ is close to one, this test mainly confirms the φ_m dependence. We therefore scanned $|C| = 2$–24 ($T = 0.48$–5.8), $\varphi_m = 0, 90°, 180°$, both backgrounds and four positions of the defect in the moiré cell (AA, AB, the wall midpoint and a generic point), with two initial states: the linear lens field and the relaxed defect-free moiré. Figure 4 shows the result. Relaxed from the lens field, the position-averaged $\langle q\rangle$ follows $T(1-\cos\varphi_m)$ over the whole range, and q equals the integer $q_{\rm lin}$ counted on the linear lens field in 88% of the runs. Relaxed from the undisturbed moiré, the defect binds fewer nodes ($q = q_{\rm lin}$ in 51%, mean $q - q_{\rm lin} = -1.5$), most strongly for $\varphi_m = 180°$ and large T. The integer bound to a core is thus a branch selected by history; the partition theorem gives the branch continuously connected to the lens field. Positional scatter is ± 0.4–2 nodes, larger for larger T. The far share is discussed in Sec. VII: relaxation screens it, so that the open symbols in Fig. 5(b) lie below $T\cos\varphi_m$.

D. Symmetry

Layer exchange ($u \to -u$, an inversion about AA) maps $(\kappa, C) \to (-\kappa, -C)$ and preserves φ_m. The mirror maps $(\kappa, C) \to (\bar\kappa, \bar C)$ and sends $\varphi_m \to -\varphi_m$. Hence (a) in the twisted moiré, removing and adding area ($\varphi_m = \mp 90°$) give identical energies and integer data; (b) in the mismatch moiré, removing and adding area are *not* related by symmetry—they bind 0 and 2 nodes; (c) the two torques in the mismatch moiré are mirror images. All three statements hold in Table I. In particular,

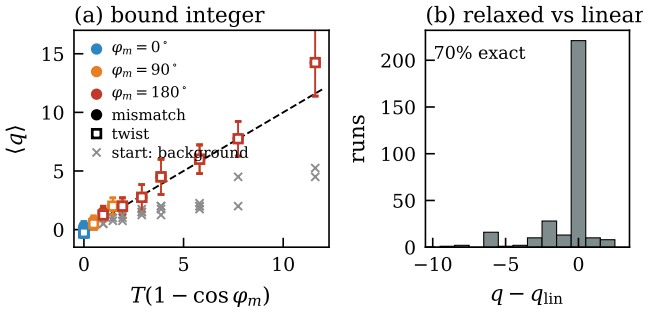


FIG. 4. Partition theorem over a range of lens strengths ($|C| = 2$–24, box half-width 72–150). (a) Bound integer q averaged over four defect positions, relaxed from the linear lens field (symbols with standard deviation; filled: mismatch, open: twist), compared with $T(1-\cos\varphi_m)$ (dashed). Crosses: the same defects relaxed from the defect-free moiré. (b) Difference between the relaxed bound integer and the integer counted on the linear lens field, all runs except AA positions with $\varphi_m = 0$ (where the linear field has an exact Einstein ring).

a closing iris and an opening iris carry opposite conserved charge and opposite far charge, yet the same sign of mean bound charge, since $1-\cos\varphi_m \geq 0$; individual cores in pairs can bind $q = -1$ (Fig. 3(b)).

V. SEVERAL DEFECTS: CONSERVATION AND ITS LIMITS

For a large loop γ enclosing all defects, write $u = u_b + \delta u$ on γ. To first order in δu, the area enclosed by the image curve changes by $\theta \oint [\cos\alpha\, \delta u \cdot n + \sin\alpha\, \delta u \cdot t] \mathrm{d}s$. Hence

$$Q(\gamma) - Q_{\rm bg}(\gamma) = -\frac{2\pi}{A_c} \mathrm{Re}\big(\bar\kappa\, C_{\rm eff}(\gamma)\big), \qquad (11)$$
$$C_{\rm eff}(\gamma) = -\frac{1}{2\pi} \oint_\gamma \delta u \cdot (n + i\,t)\, \mathrm{d}s,$$

TABLE I. Eight complex defects ($|C| = 4$) in the top layer. "Along twist" means a rotation of the top layer in the sense of the background twist. Predicted local share $T(1-\cos\varphi_m)$, measured bound integer q, and defect energy relative to the background.

defect	φ_m	pred.	q	$E - E_{\rm bg}$
mismatch, area removed	0°	0.00	0	14.34
mismatch, torque ±	±90°	0.97	1, 1	14.75, 14.75
mismatch, area added	180°	1.93	2	16.20
twist, torque against twist	0°	0.00	0	14.69
twist, area removed/added	∓90°	0.97	1, 1	15.74, 15.74
twist, torque along twist	180°	1.93	1	18.25

up to terms of order $|m|^2/\rho^2$. Multipoles $e^{-in\phi}$ with $n \geq 2$ do not change the area. If γ lies in the linear region and the moiré induced no forces, $C_{\rm eff} = \sum_k C_k$, and the far node charge would be linear in, additive in, and conserved with the atomic charge.

Figure 6(a,b) tests this on single defects and pairs in the standard box. We compare $\Delta Q(\rho)$ averaged over $36 < \rho < 58$ with the linear prediction from $\sum C$, including the image fields of the clamped box. Over 17 states the slope is 0.81 and the correlation 0.91. The integer $\Delta Q(\rho)$ jumps by ± 2 to ± 6 whenever a symmetric shell of nodes crosses the loop, so individual values carry an uncertainty of about ± 0.5.

Pairs. With $V_0 = 0$ (pure elasticity), $(+,-)$ and $(+,+)$ pairs have almost the same interaction energy: -2.99 and -3.14 at $d = 8$, $+0.06$ and -0.07 at $d = 40$ [Fig. 6(c)]. The moiré distinguishes signs only at contact. In the mismatch moiré at $d = 0.73R$ the moiré-mediated energies are $U_{+-} = -1.13$ and $U_{++} = +1.03$, and the $(+,-)$ pair binds no node. Within the lens map the removed node charge is $n_M \int \min(1, |f'|^2) \mathrm{d}^2 z$ with $f' = -\sum_k m_k/(\bar z - \bar z_k)^2$ (Appendix B). It is subadditive for opposite charges because $|C_1 + C_2| < |C_1| + |C_2|$: the attraction between a closing and an opening iris is a contact interaction of range $\sim 2R$.

In the twisted moiré, $(+,+)$ pairs show several integer branches at the same separation, binding 1, 2, or 3 nodes in total [Fig. 6(d)]. Pulling the cores apart from short distance keeps a "charge-starved" branch (cores bind $(+1, 0)$). In it, a doubled wall tethers the two cores and traps an AA node that neither core absorbs [Fig. 6(e)]. Pushing them together from large distance keeps the branch in which each core binds one node [Fig. 6(f)]. At $d = 40$ the two branches differ by 4.3 in energy. Snapping of the tether is the event in which the second core absorbs the trapped node ($q: 0 \to +1$). The twisted-moiré pair energies hardly change between $K/\mu = 0.5$ and 2: the ratio of dilational to shear strain in the twisted background is 3×10^{-4}, so the problem stays in the shear sector.

VI. THE DRESSED CHARGE

A. Branches with equal atomic charge have different far charge

To separate the far field from the box boundary we embedded the small-box branches into boxes of half-width 148.5 and relaxed them again. Two branches with the same $\sum C$ have loop charges that differ out to $\rho \approx 140$ and disagree with the bare-charge prediction. Equation (11), however, evaluated with the measured $C_{\rm eff}(\rho)$, reproduces the loop charge within 0.1–0.2 for $\rho \geq 50$ in all four states. For $\rho \gtrsim 50$, $C_{\rm eff}(\rho)$ is fitted by a point source screened by the clamped edge, $C_0(1 - \rho^2/L^2)$ with $L \approx 152$–166 [Fig. 7(a,b)]. The reorganization of the wall network is thus confined within ~ 50 of the pair, and far-

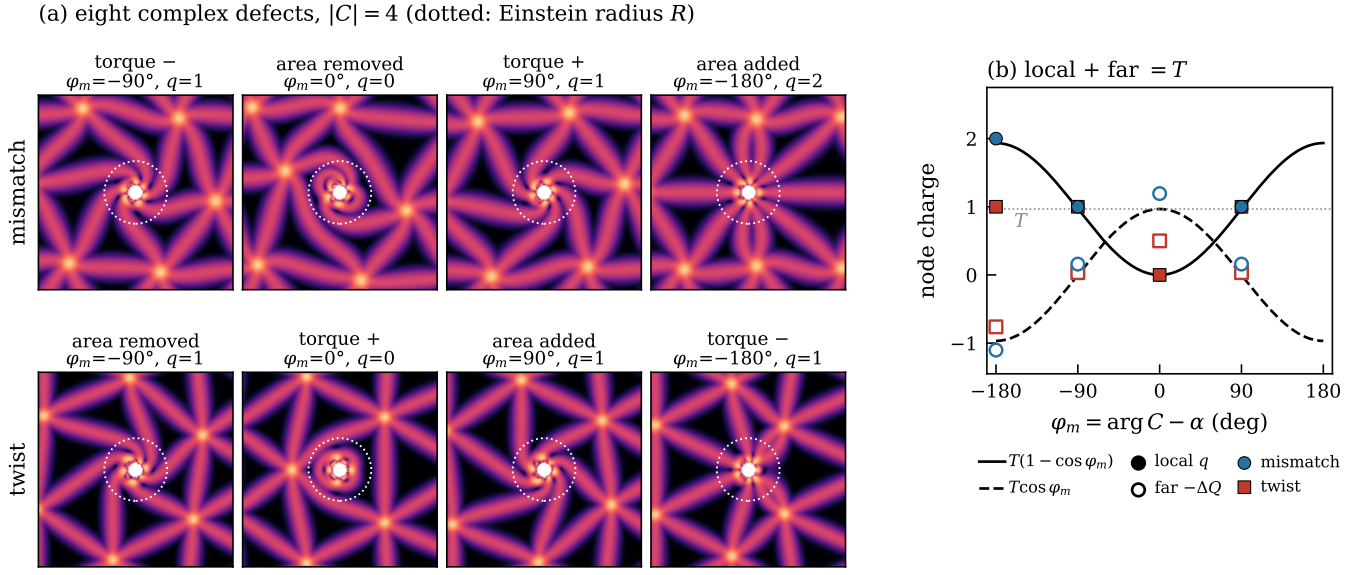


FIG. 5. Partition of the removed node charge. (a) Registry energy $\log_{10} V(u)$ around the eight defects $C = \pm 4, \pm 4i$ in the mismatch (top) and twisted (bottom) moiré, ordered by φ_m. Each title gives φ_m and the integer q bound to the core. (b) Local share q (filled) and far share $-\Delta Q$ (open, node count averaged over $36 < \rho < 58$ in a clamped box of half-width 72) versus φ_m, compared with $T(1 - \cos\varphi_m)$ and $T\cos\varphi_m$, where $T = 2\pi\theta|C|/A_c = 0.967$. Blue circles: mismatch; red squares: twist.

ther out it appears only as a monopole $C_0 \neq \sum C$.

The difference $C_{\rm reg} = C_0 - C_{\rm bare}$ is an induced charge. Here $C_{\rm bare}$ is the dressed charge of a purely elastic run with identical boundary conditions and the same fitting form. It differs from the nominal 8 (9.71 for the twisted pair, 6.69 for the mismatch pair) because the rim penalty fixes the mean rim displacement rather than the far flux, and because of the clamped-box images; using the elastic run as reference removes these effects from $C_{\rm reg}$. In the twisted moiré a torque charge appears even though the bare torque is zero, and its sign differs between branches. In the mismatch moiré the area charge differs by 3.0 between branches (Table II). The far node charges of the two branches differ by 2.7 (twist) and 0.7 (mismatch). *The far charge is therefore not fixed by* $\sum C$*; the integer branch enters through* $C_{\rm reg}$.

B. Force-dipole formula

The registry force $f = -\partial V/\partial u$ is a body force acting within the layer. When a defect reorganizes the wall network, the body force changes by Δf relative to the background. If Δf is localized and carries no net force, then in two-dimensional Lamé elasticity an isotropic force dipole $P\delta_{ij}$ produces $u_r = P/[2\pi(\lambda + 2\mu)r]$ and a point torque M produces $u_\theta = M/(4\pi\mu r)$ [27]. The traceless symmetric part produces $e^{\pm 2i\phi}$ fields that carry neither flux nor circulation. Comparing with $u_r = -\mathrm{Re}\, C/r$ and $u_\theta = -\mathrm{Im}\, C/r$ gives

$$C_{\rm reg} = -\frac{\mathrm{Tr}\, P}{4\pi(\lambda + 2\mu)} - i\,\frac{M}{4\pi\mu}, \tag{12}$$

with $P_{ij} = \int x_i\, \Delta f_j\, \mathrm{d}^2x$ and $M = P_{xy} - P_{yx}$. The moiré lattice itself carries a periodic force distribution that cancels cell by cell. The moments must therefore be evaluated with a window that decays smoothly over more than one moiré period; a sharp cutoff produces large oscillations. With windows of radii 40–100 and 50–120, Eq. (12) reproduces the measured twist torque within 0.3 and the area part within 0.6 [Fig. 7(c)].

C. Stiffness dependence of the induced charge

Equation (12) weights the area part by $1/(\lambda + 2\mu)$. If the registry forces were fixed, the area part of $C_{\rm reg}$ would halve when $\lambda + 2\mu$ doubles. We relaxed the same twisted branches at $K/\mu = 2$ ($\lambda = 1$, box half-width 109.8, with its own elastic reference). Equation (12) still holds at each stiffness, but the forces are not fixed. $\mathrm{Tr}\, P$ grows from 68 to 127 (pulled) and from 40 to 50 (pushed), and $\mathrm{Re}\, C_{\rm reg}$ changes by factors 1.07 and 0.87 instead of 0.5. The torque moments M are almost unchanged ($-92/-79$ and $+42/+49$), as expected for a reorganization confined to the shear sector [Fig. 7(d)]. The evidence comes from one separation and two branches; for the pushed branch the growth of $\mathrm{Tr}\, P$ alone would predict a factor 0.63.

The registry reorganization therefore acts more like a source of prescribed *displacement*—the partial Burgers vectors carried by the walls—not of prescribed force. $C_{\rm reg}$ is a geometric quantity selected by the integer branch, and the force dipole is the elastic reaction needed to produce it. In the mismatch case at $K/\mu = 2$ the background develops a spontaneous chirality, which makes individual

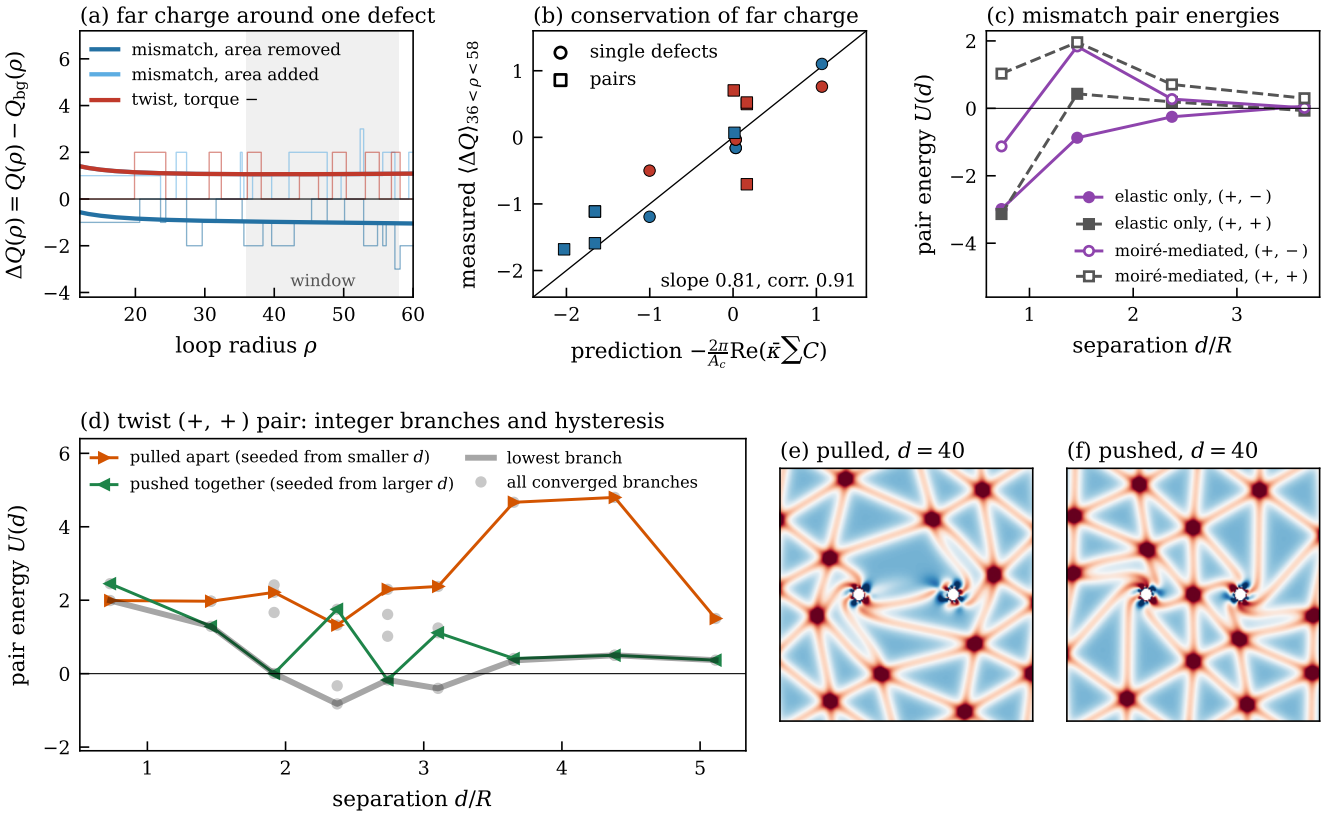


FIG. 6. Far charge and pair interactions. (a) Loop charge $\Delta Q(\rho)$ around single defects: integer data (thin steps) and linear prediction from $\sum C$ with box images (thick). The shaded window is used for averaging. (b) Measured versus predicted far charge for 8 single defects (circles) and 9 pair states (squares); blue: mismatch, red: twist. (c) Pair energy in the mismatch background: purely elastic part ($V_0 = 0$, filled) and moiré-mediated part (open) for closing–opening $(+,-)$ and closing–closing $(+,+)$ pairs. (d) Twisted $(+,+)$ pair: energies of all converged branches (gray), of the branch followed when pulling the cores apart (orange) and pushing them together (green), and the lowest branch. (e),(f) Local rotation relative to the background twist, $\omega - \theta$ with $\omega = \frac{1}{2}\mathrm{curl}\, u$, of the pulled and pushed states at $d = 40$ (red: walls and nodes, $\omega > \theta$; blue: domains, $\omega < \theta$; color range ± 0.07). In (e) a doubled wall tethers the two cores.

TABLE II. Dressed charge in large boxes ($K/\mu = 0.5$). q: integers bound by the two cores; C_0: fitted monopole; $C_{\mathrm{reg}} = C_0 - C_{\mathrm{bare}}$ measured and from Eq. (12); F: far node charge $-(2\pi/A_c)\mathrm{Re}(\bar\kappa C_0)$. In the mismatch case only the area part enters F.

state	q	C_0	C_{reg}	C_{reg} [Eq. (12)]	F
twist, pulled	(1,0)	$6.73 + 7.55i$	$-2.98 + 7.55i$	$-3.6 + 7.3i$	-1.83
twist, pushed	(1,1)	$8.11 - 3.70i$	$-1.60 - 3.70i$	$-2.1 - 3.4i$	$+0.90$
mism., pulled	(0,−1)	7.11	$+0.42$	0.0 ± 1	-1.72
mism., pushed	(0,0)	4.10	-2.59	≈ -3.0	-0.99

C_{eff} unreliable. The branch difference is still well defined: $\mathrm{Re}\,\Delta C_{\mathrm{reg}} = 2.6$ measured versus 2.55 from Eq. (12), close to the value 3.0 at $K/\mu = 0.5$.

D. Integer bookkeeping and moiré Burgers circuits

Can C_{reg} be predicted from integer data? By the Gauss law, the κ-component of C_{eff} is a node count. Consider a smooth radial window $w(r) = \mathrm{clip}[(62 - r)/32, 0, 1]$ and the window-weighted counts $N_{-,w}$ and $\Delta N_{+,w}$. Then

$$F \equiv -\frac{2\pi}{A_c}\mathrm{Re}(\bar\kappa C_{\mathrm{eff}}) = \sum_k Q_{\mathrm{hole},k} - N_{-,w} - \Delta N_{+,w}, \quad (13)$$

Here F is the far node charge ΔQ of Eq. (11); Fig. 5(b) shows $-\Delta Q$ so that the far share appears positive. Equation (13) holds within 0.07 for all 50 pair states with $d \le 40$ [Fig. 8(a)]; the residual measures the linearize-

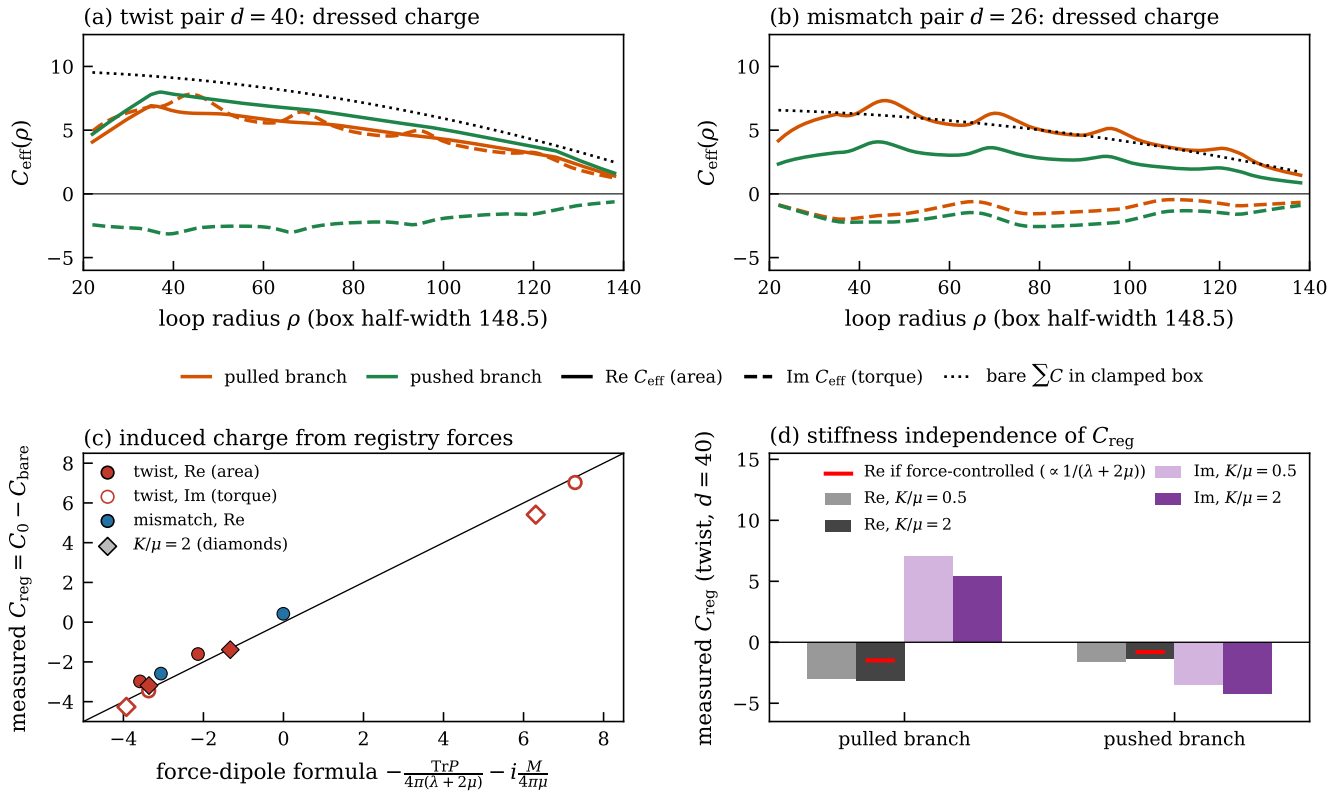


FIG. 7. Dressed charge in large boxes (half-width 148.5). (a),(b) Dressed charge $C_{\rm eff}(\rho)$ of the pulled and pushed branches of a twisted pair ($d = 40$) and a mismatch pair ($d = 26$), smoothed over one moiré period. Solid lines: area part; dashed lines: torque part. The dotted line is the bare charge in the clamped box. (c) Induced charge $C_{\rm reg}$ measured versus the force-dipole formula, Eq. (12), for $K/\mu = 0.5$ (circles) and $K/\mu = 2$ (diamonds); filled: area part, open: torque part. (d) $C_{\rm reg}$ of the twisted pair at two stiffnesses. The red bars mark the area part expected if the registry forces were fixed ($\propto 1/(\lambda + 2\mu)$).

tion and the smoothing of the window. The bound integers q_k and the core valences z_k control the sign of F in twisted pairs. With $\sum q = 1$ and one core of valence $z \le 6$, $F < 0$; with $\sum q = 1$ and both cores $z \ge 8$, or with $\sum q \ge 2$, $F > 0$ [Fig. 8(b)]. The magnitude is not fixed by (q, z), because the network deficit $\Delta N_{+,w}$ varies by up to one node between states with equal (q, z).

The missing integer is visible in moiré Burgers circuits. Since u is continuous, the moiré Burgers vector $B = A^{-1} \oint \mathrm{d}u$ vanishes for every loop, so there are no true moiré dislocations. However, labeling every node by the AA lattice point it reaches and comparing labels with the rounded geometric steps along a Delaunay triangulation reveals geometric dislocations, i.e., 5–7-like defects of the node lattice. They occur as $\pm B$ dipoles, and only in the pulled (tethered) branches [Fig. 8(c)]. Their dipole moment $B \times s$ is an inserted moiré area. In Sec. VIII we ask whether moiré elasticity converts it into far charge with a universal factor.

VII. SCREENING OF THE LENS MASS

The dressed charge of Sec. VI acts already on a single defect. We define the far lens mass $m_{\rm eff} = C_{\rm eff}/\kappa$ and the screening ratio $\rho = m_{\rm eff}/m$. To measure it we track every relaxed node by its label and fit its displacement from the defect-free position to the lens term $m_{\rm eff}/\bar{w}$, evaluated at the relaxed position, plus a shear dipole, the Kelvin terms of a net core force, and the linear terms induced by the clamped boundary. The fit is independent of the initial state to within 0.01 and converges with box size (half-widths 150 and 200 agree within 0.01); the boundary image term decays as $1/L^2$. Node counts on circles, used in Fig. 5(b), are biased by the clamped boundary and serve only as a check.

An exact identity. For the twisted background and $\varphi_m = 0$ the defect field is tangential. Let $\delta\omega$ be the local rotation in excess of the background twist. Stokes' theorem applied between the core rim, where the circulation is fixed by C, and infinity gives

$$|C_{\rm eff}| = |C| - \frac{1}{\pi}\int \delta\omega \, \mathrm{d}^2 r, \qquad T(1-\rho) = \frac{2\theta}{A_c}\int \delta\omega \, \mathrm{d}^2 r. \tag{14}$$

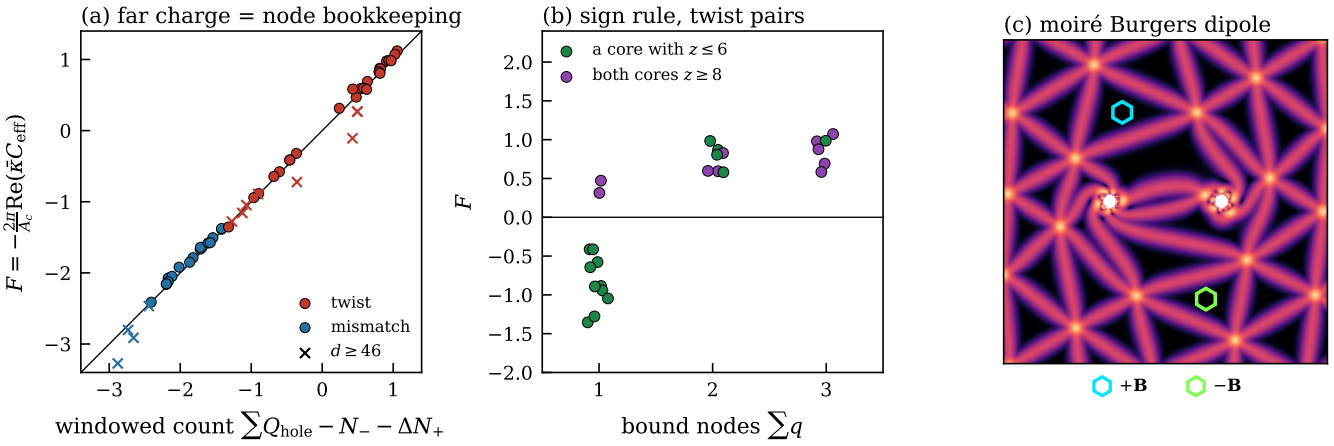


FIG. 8. Integer bookkeeping. (a) Far charge F versus the window-weighted integer count of Eq. (13) for 53 pair states (crosses: $d \geq 46$, cores close to the window edge). (b) Sign rule for twisted pairs ($16 \leq d \leq 46$): F versus the total bound charge $\sum q$, colored by the smallest core valence. (c) Geometric moiré Burgers circuits in a pulled twisted pair ($d = 40$; color $\log_{10} V$): hexagons mark Delaunay triangles of nodes with a non-closing circuit; their $\pm B$ form a dipole.

The screened part of the far share is therefore exactly the number of extra nodes created by over-twisting the moiré near the defect. For the mismatch background the same holds with the excess half-dilation in place of $\delta\omega$. Since the pure-shear field $-C/\bar{z}$ solves the equations of any homogeneous linear moiré elasticity ($\partial_z \delta u = 0$), no choice of moiré moduli can produce $\rho \neq 1$: screening must come from the nonlinear zone $r \lesssim R$.

Measurements. Figure 9 summarizes 96 relaxed defects ($|C| = 8$, $\varphi_m = 0$, box half-width 150). Without registry, $\rho = 0.96$–0.99 (0.89 for the softest bulk, $K/\mu = 0.1$). In the twisted moiré ρ decreases monotonically with the relaxation strength $V_0/\mu\theta^2$ and with the bulk stiffness: at $V_0 = 0.005$, $\rho = 0.52$, 0.41, 0.35, 0.29 for $K/\mu = 0.1$, 0.5, 1, 2, and at $V_0 = 0.02$, $\rho = 0.10$–0.20. The ratio is independent of $|C|$ ($|C| = 4, 8, 16$ agree within 0.04) and of the defect position, except for a defect on an AA site, which screens about twice as strongly. The mismatch moiré screens less: for $K/\mu \leq 0.5$ the lens is even slightly amplified at weak registry (ρ up to 1.2) and screened at strong registry ($\rho = 0.2$–0.25 at $V_0 = 0.02$). For $K/\mu \geq 1$ and weak registry the whole mismatch moiré rearranges around the defect and ρ is not well defined in our boxes. The far share $-\Delta Q \approx 0.5\,T$ of the twisted torque in Fig. 5(b) is the box-biased version of $\rho = 0.41$.

Mechanism. The relaxed core is surrounded by a rosette of curled walls inside about $R/\sqrt{2}$, beyond which the moiré is nearly undisturbed [Fig. 9(a,b)]. Inside the rosette the local twist exceeds the background by $\bar{b} \approx (0.3$–$2)\theta$, growing with registry strength. If this over-twisted disk were uniform and joined the outer moiré at its own Einstein radius, continuity of u would give $m_{\rm eff} = C/\kappa_{\rm in}$ with $\kappa_{\rm in} = i(\theta + \bar{b})$, i.e., the far moiré sees the defect through the local twist of the disk. This estimate, $\rho \approx \theta/(\theta + \bar{b})$, matches for weak pinning ($\rho \gtrsim 0.6$) but overestimates ρ for strong pinning, where Eq. (14) requires additional rotation spread over $r \lesssim R$.

Local-density theory. The cost of over-twisting is set by the energy density $e(G)$ of a relaxed moiré with a uniform deformation gradient G, which we computed in periodic cells for a background twist or mismatch θ plus a shear η [Fig. 9(c)]. Its registry part is nearly constant for $|\eta| < \theta$ and rises toward the unrelaxed value $3V_0$ once the moiré is inverted ($|\eta| > \theta$), because a dense inverted moiré can no longer form narrow walls. Near a defect $|\eta| = |C|/r^2 > \theta$ for $r < R$, so the whole Einstein disk is in this expensive state. Over-twisting the disk raises the local twist $\theta + \delta\omega$ and thus lowers the ratio $|\eta|/(\theta + \delta\omega)$ that controls the registry energy, while the rotation itself costs no elastic energy; the only elastic cost is the shear $|\delta\omega + A/r^2|$, where $A(r) = |C| - 2\int^r \delta\omega\, s\, \mathrm{d}s$ follows from Eq. (14). Minimizing $\int e(G)\, \mathrm{d}^2 r$ over radial profiles $\delta\omega(r)$ within the inverted zone, with the background pre-stress removed by the constraint of a fixed far boundary, gives the bands in Fig. 9(d). Apart from $e(G)$, the only input is the radius of the inverted zone, taken between $0.75R$ and R. The theory reproduces the decrease of ρ with registry strength and the size of the over-twist [Fig. 9(e); within 35% for moderate pinning, up to a factor 1.7 too large for the strongest]. It also reproduces the opposite stiffness trends. In the twisted moiré a stiffer K makes sheared moirés more expensive [Fig. 9(c)], because sheared walls acquire a tensile component, so screening grows with K. In the mismatch moiré the over-dilation needed for screening costs K, and a weak shear turns tensile walls into cheaper mixed walls, which favors a slight amplification. Being local, the theory lacks the gradient energy of walls on the scale L_M and overestimates screening at strong pinning.

Phase dependence. For $\varphi_m = 0$ the screened state is unique. For $\varphi_m = 180°$ and 90° we found two families of states with different near-core structure. In one,

TABLE III. Moiré elastic constants in units of θ^2 ($\lambda = -0.5$, $\mu = 1$). The last row gives the values without registry relaxation (twist/mismatch).

	K_M	μ_M	χ_M	σ_d	$K_M/(K_M+\mu_M)$
twist	−0.149	1.149	6.41	−3.34	−0.149
mismatch	0.273	0.479	−1.29	−5.92	0.363
bare	0/0.5	1/1	2/0	–	0/0.33

the inner disk is nearly unchanged (in the twisted moiré $\rho \approx 0.9$ for $\varphi_m = 180°$). In the other, the disk undertwists as Eq. (14) requires for a negative mass, and at strong registry its local twist reverses sign, which screens the lens almost completely ($\rho \approx 0.04$). Which family is the ground state is not settled by our clamped-boundary calculations. For $\varphi_m = 90°$ the screening is complex, $m_{\rm eff}/m = \rho e^{i\delta}$ with $|\delta| \lesssim 25°$ and opposite signs of δ in the two families, so the lens is also rotated.

VIII. MOIRÉ ELASTICITY AND THE RING-TRACTION IDENTITY

A. Elastic constants of the moiré crystal

We define the Eulerian moiré displacement U by $u = \kappa(z-U)$ and impose homogeneous strains $U = U_a z + U_b \bar{z}$ on oblique periodic cells spanned by the strained moiré vectors (2×2 moiré cells). We then fit the relaxed energy density as

$$e = e_0 + \sigma_d \varepsilon_d + 2K_M \varepsilon_d^2 + \tfrac{1}{2}\chi_M \omega^2 + 2\mu_M |\gamma|^2 + \dots, \quad (15)$$

with $\varepsilon_d = \mathrm{Re}\, U_a$, $\omega = \mathrm{Im}\, U_a$ and $\gamma = U_b$ (35 strain states, residual $< 10^{-8}\theta^2$). All terms forbidden by hexagonal symmetry—linear shear, anisotropy, and the $\varepsilon_d\omega$ cross term—vanish within numerical precision. The results (Table III, Fig. 10) differ qualitatively from the unrelaxed values. A moiré dilation is an atomic rotation in the twisted case and an atomic dilation in the mismatch case, which fixes the bare entries.

Two signs stand out. The twisted moiré has a *negative bulk modulus*: a uniform moiré dilation (a local untwisting) lowers the energy. It satisfies strong ellipticity ($\mu_M > 0$, $K_M + \mu_M = 1.000\,\theta^2 > 0$) and is therefore stable under clamped boundaries, but unstable to homogeneous dilation when the boundary is free. That $K_M + \mu_M$ equals the unrelaxed value is a numerical observation we cannot yet explain. The mismatch moiré has a *negative rotational stiffness*: it tends to rotate spontaneously, consistent with the small global rotation observed in mismatch backgrounds. In clamped boxes the bulk moiré expands uniformly, $\varepsilon_0 \approx \Delta^*/L$ with $\Delta^* \approx 2.7$–3.0 for the twist [Fig. 10(c)]. The excess twist is stored in an edge layer near the clamp. Treating the bulk as uniform and the edge layer as a line energy $g(\Delta)$ of the mismatch $\Delta = (\varepsilon_b - \eta)L$, the equilibrium condition $g'(\Delta) = -\sigma_d/2 - 2K_M\varepsilon_b$ is dominated by σ_d. This explains the $1/L$ law. The edge is nevertheless stiff: when a dilation η is imposed on the clamp, the bulk follows it almost one to one [Fig. 10(d)].

B. A local moiré source

If these constants governed local sources, a disk of inserted moiré area $\delta A^* = 2\pi r_0^2 \varepsilon^*$ would send the fraction $S = K_M/(K_M+\mu_M)$ of it to infinity, in analogy with an Eshelby inclusion [26]. The far factor would then be $f = 1 - S$: 1.15 for the twist and 0.64 for the mismatch. To test this we construct the moiré analog of an eigenstrain inclusion. Inside a disk D of radius r_0 centered at z_0, the energy is evaluated at $u + s$ instead of u, with $s = \varepsilon^*\kappa(z - z_0)\mathbb{1}_D$. The registry potential becomes $V(u+s)$ and $\varepsilon^*\kappa$ is added to the elastic gradient. The natural state inside is a moiré dilated by ε^*, and the jump of s at ∂D is a ring of fractional moiré dislocations.

The far response, measured from exact node positions, does not follow S. In twisted and mismatch moirés with $r_0 = 45$–95 (1.5–3.2 moiré periods) in boxes of half-width 150 and 200, $S = A/(\varepsilon^* r_0^2)$ ranges from -8 to $+5.5$. Here A is the coefficient of the far field $A(1/r - r/L_{\rm eff}^2)$, whose effective radius $L_{\rm eff}$ always matches the box. The value of S depends on r_0, on whether the ring passes through a node shell, and on the sign of ε^* [Fig. 11(d)].

C. Ring-traction identity

The drive of such a source obeys an exact identity. Let u_0 be the equilibrium at $\varepsilon^* = 0$ and $w = \kappa(z - z_0)$. By the envelope theorem,

$$E_1 \equiv \frac{\partial E}{\partial \varepsilon^*}\Big|_0 = \int_D \sigma_{\rm at} : \nabla w \,\mathrm{d}A + \int_D \nabla V(u_0)\cdot w \,\mathrm{d}A,$$

where $\sigma_{\rm at}$ is the atomic Cauchy stress of u_0. The equilibrium condition $\nabla V(u_0) = \mathrm{div}\,\sigma_{\rm at}$ and one integration by parts cancel the bulk terms exactly:

$$E_1 = \oint_{\partial D} \left(\sigma_{\rm at}\, n\right)\cdot \kappa\,(z - z_0)\,\mathrm{d}l. \quad (16)$$

The local source therefore couples only to the microscopic traction on its boundary [Fig. 11(a)]. For the twist, $\kappa(z - z_0)$ is tangential and $E_1 = \theta r_0 \oint t_t \,\mathrm{d}l$ is the microscopic torque on the ring. Averaged over the position z_0 within a moiré cell, the mean stress is symmetric and $\langle E_1\rangle = 0$. For the mismatch, $\kappa(z - z_0)$ is radial and $E_1 = \theta r_0 \oint t_n \,\mathrm{d}l$. The mean atomic stress produced by the background strain θ gives $\langle E_1\rangle = 4(\lambda+\mu)\theta^2\pi r_0^2$, which equals $2\theta^2\pi r_0^2$ here. A continuum moiré description would instead predict $E_1 = -\pi r_0^2\sigma_d$ (+57 and +101 at $r_0 = 70$ for twist and mismatch). This coupling is absent. The mismatch mean exists because the total-u

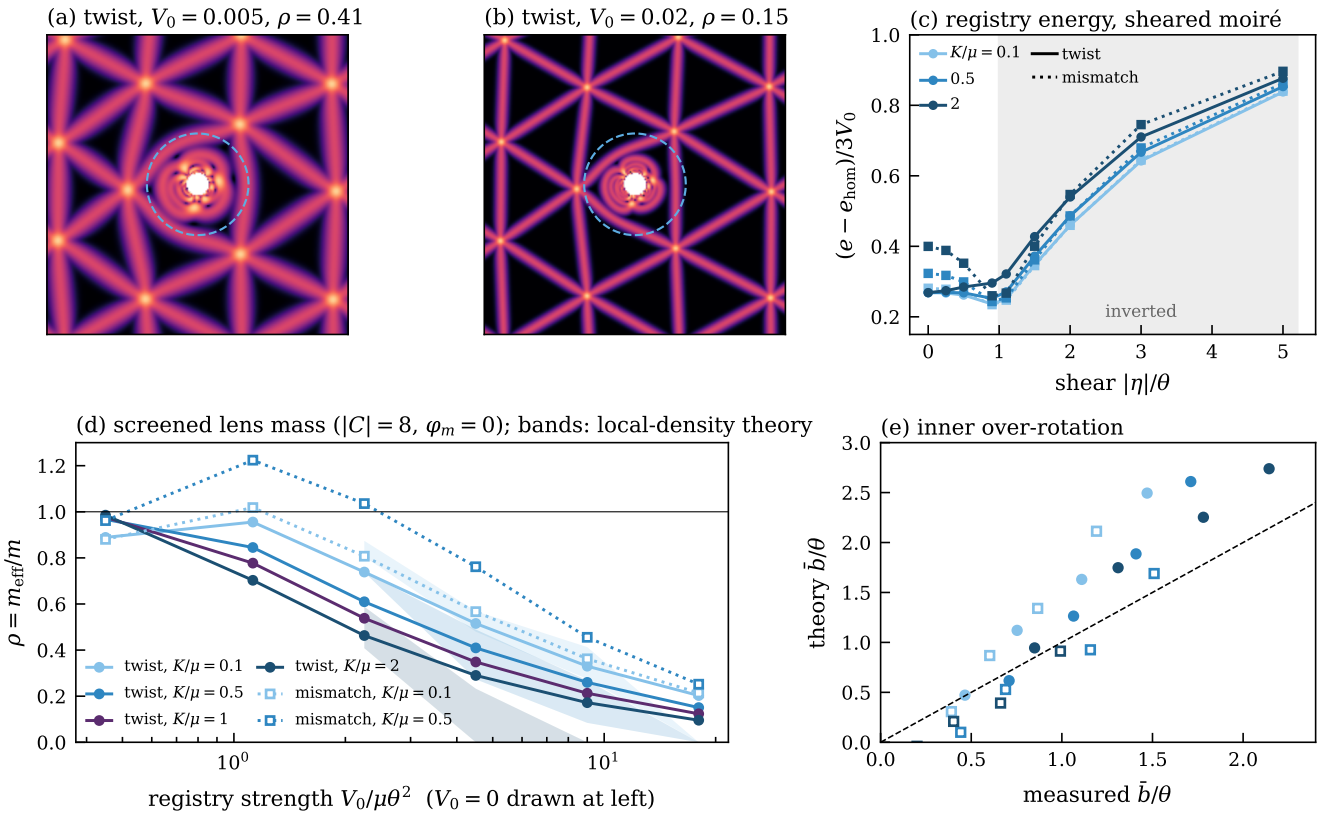


FIG. 9. Screening of the lens mass. (a),(b) Relaxed twisted moiré around a torque defect ($|C| = 8$, $\varphi_m = 0$, $K/\mu = 0.5$) for two registry strengths, $\log_{10} V$; dashed: Einstein radius. Stronger registry confines the distortion to a rosette and leaves the outer moiré nearly undisplaced. (c) Registry part of the energy density of a relaxed moiré with background θ plus uniform shear η ($V_0 = 0.005$); shading: inverted moiré ($\det G < 0$). (d) Screening ratio $\rho = m_{\rm eff}/m$ from tracked node displacements versus registry strength (box half-width 150). Shaded bands: local-density theory for twist ($K/\mu = 0.1$, 0.5, 2; band edges: cutoff $0.75R$ and R). (e) Mean over-twist $\bar{b}$ of the inner disk ($r < 0.6R$): local-density theory versus simulation (circles: twist, squares: mismatch; colors as in (d)).

reference stresses the mismatch background; in a stress-free reference it would vanish as well.

Table IV and Fig. 11(b,c) verify Eq. (16). Evaluated on the baseline field alone, it reproduces the finite-difference drive of the inclusion runs within 1–2% in the linear regime ($|\varepsilon^*| \le 0.01$). Averages over 16 ring centers agree with the predicted means within one standard error. The fluctuation between placements grows faster than r_0, with a standard deviation of about 0.3 of the mismatch mean, and the centered, sixfold-symmetric ring is always an extreme. The negative-ε^* jumps seen in the mismatch runs are rearrangements of nodes next to the ring, where the drive is concentrated.

D. Consequence

Equation (16) concerns the drive, not the response: the far coefficient S also depends on how the surrounding moiré responds, and for $r_0 \le 3.2L_M$ no homogenized description is expected. The identity nevertheless explains why S scatters with placement and why it does not follow $K_M/(K_M + \mu_M)$. The placement-averaged drive is universal, but it is set by the atomic background stress rather than by the moiré pre-stress σ_d. The same reasoning applies to every moiré-scale defect embedded in a continuous registry, including the geometric $\pm B$ dipoles

TABLE IV. Ring-traction identity, Eq. (16). E_1 from the ring integral on the baseline field, from finite differences of the inclusion runs, and the mean ± standard deviation over 16 ring centers, compared with the predicted mean.

	r_0 (box)	identity	runs	16 centres	pred.
twist	45 (150)	−8.82	−8.80	-0.4 ± 2.9	0
twist	70 (150)	−16.94	−16.84	-1.7 ± 9.0	0
mism.	45 (150)	0.88	0.90	14.1 ± 4.2	14.1
mism.	60 (150)	33.10	32.6	26.5 ± 7.4	25.1
mism.	70 (150)	7.65	7.76	32.2 ± 11.2	34.2
mism.	95 (200)	21.99	–	60.8 ± 21.2	63.0

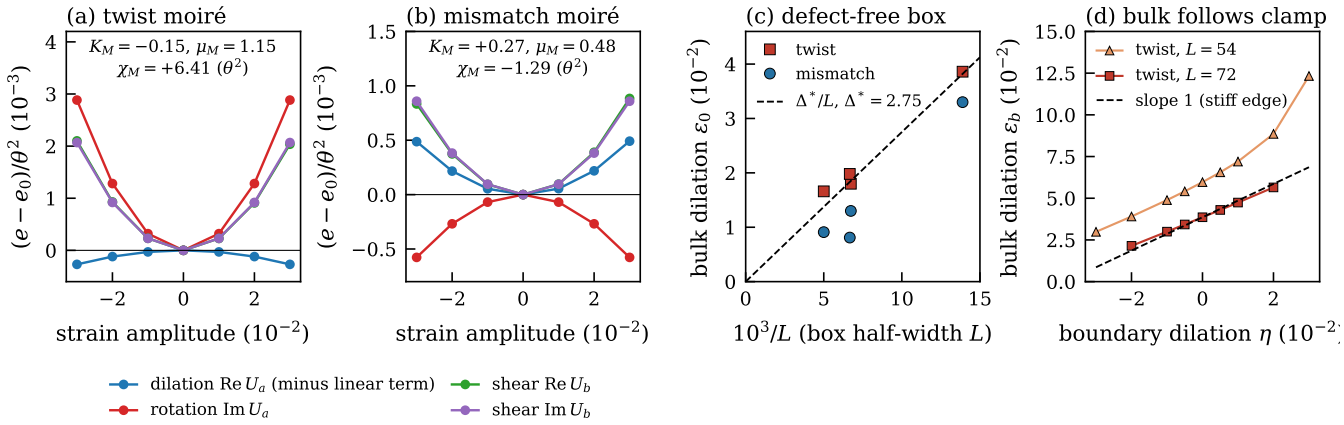


FIG. 10. Elasticity of the moiré crystal. (a),(b) Relaxed energy density versus homogeneous moiré strain for the twisted and mismatch moiré (periodic cells). The linear pre-stress term $\sigma_d \varepsilon_d$ has been subtracted from the dilation curve; the fitted constants are given in units of θ^2. (c) Uniform bulk dilation ε_0 of defect-free clamped boxes versus inverse box half-width. (d) Bulk dilation of twisted boxes versus a dilation η imposed on the clamped boundary; the dashed line has slope one.

of Fig. 8(c): they couple through the forces their cores exert on the atomic layer. For the far node charge we therefore rely on the exact accounting of Eq. (13) together with the force-dipole dressing of Eq. (12); this is an accounting, not yet a predictive theory of $C_{\rm reg}$. Whether a placement-averaged S converges for $r_0 \gg L_M$ remains open. The moiré constants of Table III describe homogeneous strains, such as the boundary-driven and $1/L$ bulk dilations of Fig. 10.

IX. LENSES LARGER THAN THE MOIRÉ PERIOD

The simulations above have $R \approx 11 < L_M \approx 30$ and $T = 0.97$, at the boundary of regimes I and II defined below; each defect removes about one node. What happens when the lens is larger than the moiré lattice constant? Three lengths enter. The atomic core $r_c \simeq \sqrt{|C|}$ is where the defect strain $|\nabla \delta u| = |C|/r^2$ reaches order one. The Einstein radius $R = \sqrt{|C|/\theta}$ is where the defect strain equals the background registry gradient θ. The moiré period is $L_M = a/\theta$. Their ratios are

$$\frac{r_c}{R} = \sqrt{\theta}, \qquad \frac{R}{L_M} = \sqrt{\frac{T A_c}{2\pi}}, \qquad T = \theta n_{\rm miss}. \tag{17}$$

Since the atomic strain at the Einstein ring equals $\theta \ll 1$, linear elasticity of the layer, and hence the lens map, holds everywhere outside r_c regardless of the lens size. Large lenses are in fact the cleanest. Equation (17) defines four regimes [Fig. 12(d)]: (I) $T < 1$, where the far distortion is a fraction of a node; (II) $1 < T < 2\pi/A_c \approx 7$, the few-node lens with $R < L_M$; (III) $R > L_M$ but $r_c < L_M$, the resolved lens; and (IV) $n_{\rm miss} > 2\pi/(A_c\theta^2)$, where the core itself exceeds the moiré period.

Arcs and the moiré Einstein ring. In regime III the critical circle $|z| = R$, where $J = 0$, is resolved by many nodes. It maps onto the caustic $|\beta| = b_s$ of the source plane. Node images close to it are stretched tangentially into arcs, exactly like lensed galaxies [Fig. 12(a)]. When $\varphi_m = 0$ the caustic shrinks to a point. On $|z| = R$ one has $\beta = z - R^2/\bar{z} = 0$, so the entire circle carries the single registry value of the undeformed pattern at the defect position. The result is a closed ring of uniform stacking, a *moiré Einstein ring* [Fig. 12(b)]. Its stacking is AA if the defect sits on an AA site and AB if it sits on an AB site, so the ring images the registry at the defect position. In a mismatch bilayer, removing area from the larger-lattice layer (or adding area to the smaller-lattice layer) gives $\varphi_m = 0$ and such a ring once $T \gtrsim 7$; the opposite combinations give $\varphi_m = 180°$ and swallow the maximal disk $b_s = 2R$. In a twisted bilayer a ring requires $\varphi_m = 0$, i.e., a torque charge; an area defect produces arcs and swallows the nodes inside $b_s = \sqrt{2}R$.

Inverted moiré inside the ring. Inside R every source has an inverted image, so the region $r_c < |z| < R$ hosts a mirror-image moiré. Its node density $n_M R^4/|z|^4$ grows toward the core, and its total node number $\pi n_M R^4 (r_c^{-2} - R^{-2})$ is of order $\pi|C|/A_c = n_{\rm miss}/2$, independent of θ (the prefactor depends on the core cutoff). This inner pattern is the moiré generated by the defect's own strain field. Outside the caustic disk, the swallowed number $\pi n_M b_s^2 = T(1 - \cos\varphi_m)$ becomes large. For the linear lens map the lattice-point counting error scales with the perimeter, so the relative rounding error decreases as L_M/b_s. Relaxation, however, can shift the bound integer by more than this (see below).

Cores larger than the moiré period. In regime IV the defect contains its own moiré. A grain of radius r_g rotated by ϕ has $C = -i\phi r_g^2$ outside and a uniform local twist $\theta + \phi$ inside, so $R = r_g\sqrt{|\phi|/\theta}$. For $|\phi| < \theta$ the Ein-

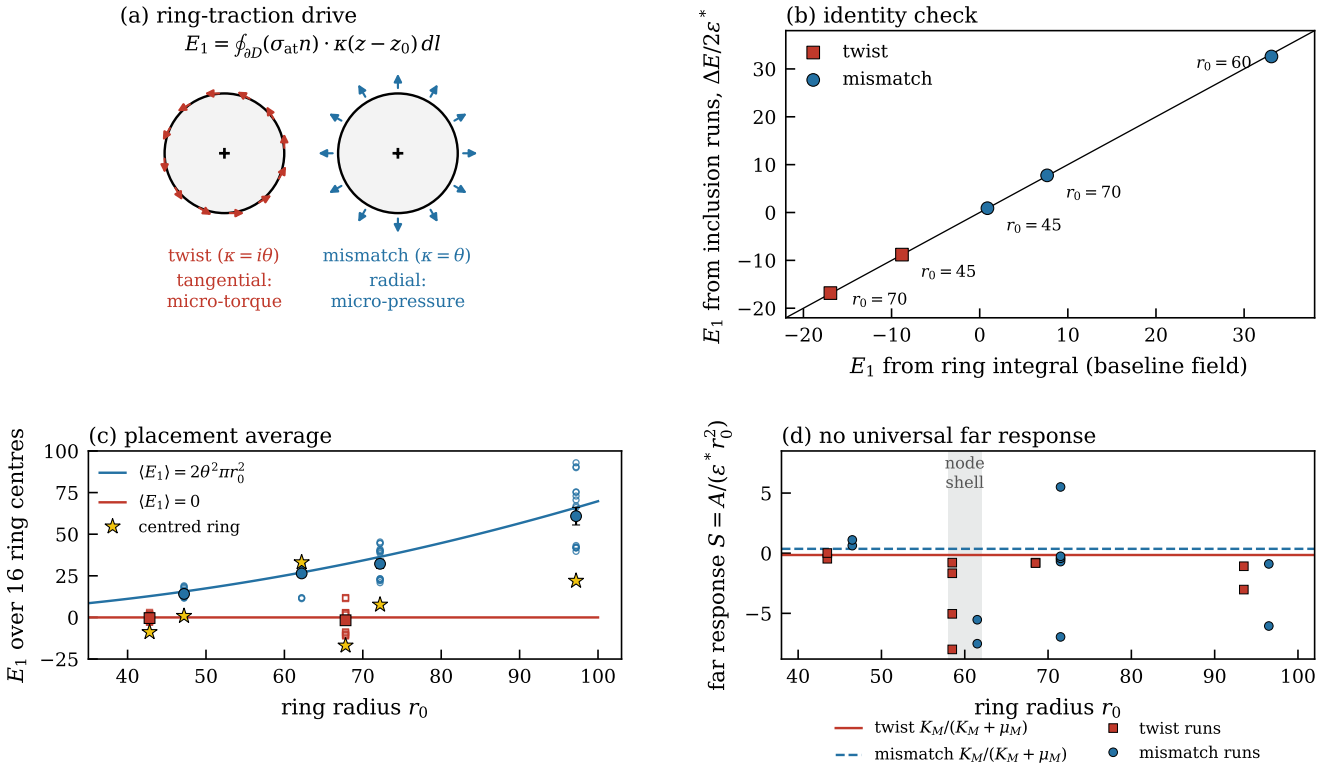


FIG. 11. Ring-traction identity. (a) The drive of a local moiré source is the atomic traction integrated against $\kappa(z - z_0)$ on its boundary: tangential (micro-torque) for the twist, radial (micro-pressure) for the mismatch. (b) E_1 from Eq. (16) on the baseline field versus the finite-difference drive of the inclusion runs. (c) E_1 for 16 ring centers per radius (small symbols), their mean ± standard error, and the centered ring (stars), compared with the predicted means 0 (twist) and $2\theta^2\pi r_0^2$ (mismatch). (d) Far response $S = A/(\varepsilon^* r_0^2)$ of all inclusion runs (single placements). No common value emerges, in particular not the Eshelby value $K_M/(K_M + \mu_M)$ (horizontal lines). The gray band marks a ring lying on a node shell.

stein circle lies inside the grain and the grain is a smooth twist-angle domain. For $|\phi| > \theta$ the critical circle lies outside the grain. In particular, a grain rotated by $\phi = -2\theta$ has local twist $-\theta$ inside and $\varphi_m = 0$, and is surrounded by an Einstein ring at $R = \sqrt{2}\, r_g$ [Fig. 12(c)]. This assumes a grain whose rotation is imposed and elastically transmitted to the surroundings. A grain bounded by a relaxed low-angle grain boundary has no long-range field, and the smooth twist-angle variations of real samples [14] are not discrete grains. Where the exterior field is $-C/\bar{z}$, however, the node accounting of Eq. (10) applies directly.

Relaxation. Figure 12(e,f) shows relaxed continuum states for the defect of panels (a,b) ($|C| = 120$, $r_h = 20$, box half-width 150). Relaxation keeps the lens structure but converts it into domain walls, and the two backgrounds relax differently.

In the twisted bilayer, the lens map places ordinary images down to the critical circle $r = R = 60$, where they are compressed into arcs. After relaxation the nearest ordinary node lies at $r \approx 92 \approx 1.5R$: the arcs dissolve, and a fan of spiral walls connects the core to the outer network. The core binds $q = Q_{\rm hole} - N_- = 37$ nodes, compared with the lens value $T = 29$. This is the lowest-energy of three states we found: relaxed from the defect-free moiré the core binds only 7 nodes (energy higher by 49), and with the defect centered on an AB site it binds 28 (higher by 10). For such large lenses the bound integer thus depends on the relaxation path by ±20 nodes. Every circle in the node-free annulus crosses exactly $z = 42$ walls. If the 37 enclosed AA points carry unit winding, Eq. (6) requires $z \geq 2L_{\rm min}(37) = 42$, since a compact hexagon of 37 lattice points meets 21 lattice lines. The relaxed large lens attains exactly this minimum. The extra bound nodes appear as dressed far charge, $\Delta Q(\rho) \approx$ 6–12 at $\rho = 100$–130 instead of the unscreened linear 3–5. Since the box half-width is only $2.5R$, this comparison is semi-quantitative, and each case is a single relaxed state.

In the mismatch bilayer the AA Einstein ring costs registry energy and does not survive. The ring and the inverted inner moiré reconstruct into nested closed domain walls bounded by the Einstein circle, i.e., a target pattern of commensurate annuli. Here the core binds $q = 1$ (lens value 0), and the ordinary-node deficit follows the lens prediction ($\Delta N_+ \approx 17$–21 against 18–26,

TABLE V. Defect hierarchy under $u = \kappa U$.

level	atomic	moiré	in own units
disclination	$s \in 60^\circ\mathbb{Z}$	same s	integer
dislocation	$b \in \Lambda$	$B = A^{-1}b \in \Lambda_M$	integer
area/torque	C (complex)	$m = C/\kappa$	$\times\theta$ (fractional)

the deficit falling short by 2–5 at the largest radii). Thus φ_m selects the relaxed texture: a lens that rotates its images ($\varphi_m = \pm 90^\circ$) relaxes into a spiral wall fan, and a lens that only dilates them ($\varphi_m = 0$) relaxes into concentric walls. When the defect sits on an AB site, the Einstein ring is already a low-energy stacking and should survive relaxation as a commensurate annulus; we have not computed this case.

Transmitted charge. A vacancy cluster lenses only through the part of its area that the surrounding layer actually closes. A large hole with free edges transmits little C, whereas inserted islands and rotated grains always transmit their full charge.

X. DISCUSSION

Hierarchy of charges. The affine map $u = \kappa U$ relates atomic and moiré defects level by level (Table V). Disclinations and dislocations are homotopy invariants ($\pi_1(T^2) = \mathbb{Z}^2$ and the point group). Because A maps lattice to lattice, an integer atomic invariant becomes an integer moiré invariant. The atomic area charge is quantized by the number of atoms but is not a homotopy invariant ($\pi_2(T^2) = 0$), whereas the moiré node number is a degree. The continuous parameter θ sits in between, so the far share $\theta n_{\mathrm{miss}} \cos\varphi_m$ is fractional, and the integer local share of the nonlinear core makes up the difference. Writing the node density as the Jacobian $(1/4\pi^2)\epsilon^{ab}\partial_a\phi_1\partial_b\phi_2$ of the two moiré phases $\phi_i = \boldsymbol{G}_i \cdot u$ makes the analogy with Goldstone–Wilczek induced charge explicit [28].

Twist versus mismatch. The two backgrounds differ only in α, but the consequences are distinct. (i) Area defects are swallowed with $b_s = \sqrt{2}R$ in a twisted moiré and not at all in a mismatch moiré. Frustrated integer assignment in the twisted case produces multiple branches and hysteresis. (ii) Relaxation of the twisted moiré is rotational and its walls are shear walls; mismatch relaxation is dilational. The bulk modulus nevertheless matters in both, through the tensile component of sheared walls, and it shifts the screening of Sec. VII in opposite directions. (iii) The mismatch background is mirror symmetric, so node rotation is an Ising-like variable, and we observe spontaneous chirality of the nodes at $K/\mu = 2$. In the twisted background the mirror is absent and node rotation is pinned.

Heterostrain. A uniform heterostrain adds $\eta\bar{z}$ to the background, $u_b = c_{\mathrm{AA}} + \kappa z + \eta\bar{z}$. The lens map becomes $\beta = z + (\eta/\kappa)\bar{z} - m/\bar{z}$, the point-mass lens with external shear (the Chang–Refsdal lens [35]). Its caustic is an astroid instead of a circle, and regions with four images appear. The relevant parameter is $|\eta/\kappa| \sim \epsilon_h/\theta$ for a heterostrain ϵ_h, so even a heterostrain of 0.1% gives $|\eta/\kappa| \approx 0.05$ near the magic angle, and the lens is distorted more strongly at smaller θ. We relaxed twisted lenses with $|\eta/\kappa| = 0.1$ and 0.3 in two shear directions ($|C| = 4$ and 24). The φ_m dependence of the bound integer survives (for $|C| = 24$, $\varphi_m = 180^\circ$: $q = 11$–14 against 11.6), the dependence on the shear direction is at most one node, and the far share of the twisted torque grows with heterostrain, from $0.48T$ to 0.6–$0.7T$ at $|\eta/\kappa| = 0.3$ (node counts, same initial state), i.e., heterostrain partly lifts the screening.

Experimental consequences. The lens scales follow from Eq. (17) without model input. A defect that removes T nodes has $R/L_M = \sqrt{TA_c/2\pi} \approx 0.37\sqrt{T}$, independent of θ; for $T = 1$ it must miss $1/\theta$ unit cells (about 52 at $\theta = 1.1^\circ$). A single removed node is therefore resolved as a lens only when $0.37L_M$ exceeds the domain-wall width, which in our model ($w \approx 0.10L_M$) is satisfied. In real bilayers the wall width is fixed in nanometres [7], so the condition favors small twist angles. Independently of resolving the swallowed region, the lens displaces the surrounding nodes by $|m|/r = R^2/r$; at $r = L_M$ this is $0.14TL_M$, about 1.8 nm per removed node at $\theta = 1.1^\circ$, which is within reach of drift-corrected STM or interferometric moiré metrology [14, 15, 17, 18]. Such a measurement yields the screened mass ρm; since ρ falls with the relaxation strength $V_0/\mu\theta^2$, the same defect should appear lighter at smaller twist angles, where relaxation is stronger. The resolved regime $R > L_M$ requires $n_{\mathrm{miss}} > 2\pi/(A_c\theta) \approx 7.3/\theta$ unit cells, about 380 at 1.1°. A real vacancy cluster lenses only with the charge its surroundings actually close; multivacancy reconstruction and buckling reduce C below $n_{\mathrm{miss}}A_c/2\pi$ [20]. Our mismatch case uses a homobilayer registry with degenerate AB and BA stackings. A heterostructure such as graphene on hBN has inequivalent stackings, so our predictions for it are qualitative: an area defect lenses with $\varphi_m = 0$ or 180° depending on the host layer, and whether the Einstein ring of a $\varphi_m = 0$ lens survives relaxation depends on the energy of the stacking it carries. Finally, two defects with the same atomic content can differ in far node charge by up to ~ 3 nodes in our simulations (one pair geometry), so the moiré around a defect carries information about its core state.

Model assumptions. The topological statements (Secs. III and IVA) hold for any continuous registry map with vanishing Burgers class. The partition theorem relies on the linear region outside the core, i.e., on R being larger than the wall width. It also requires that the registry forces produce no net stress on average over the moiré lattice. The single relative-displacement field carries the stiffness of the bilayer combination; how a defect's field is shared between the layers rescales C but not the lens structure. The elastic parameters, in particular

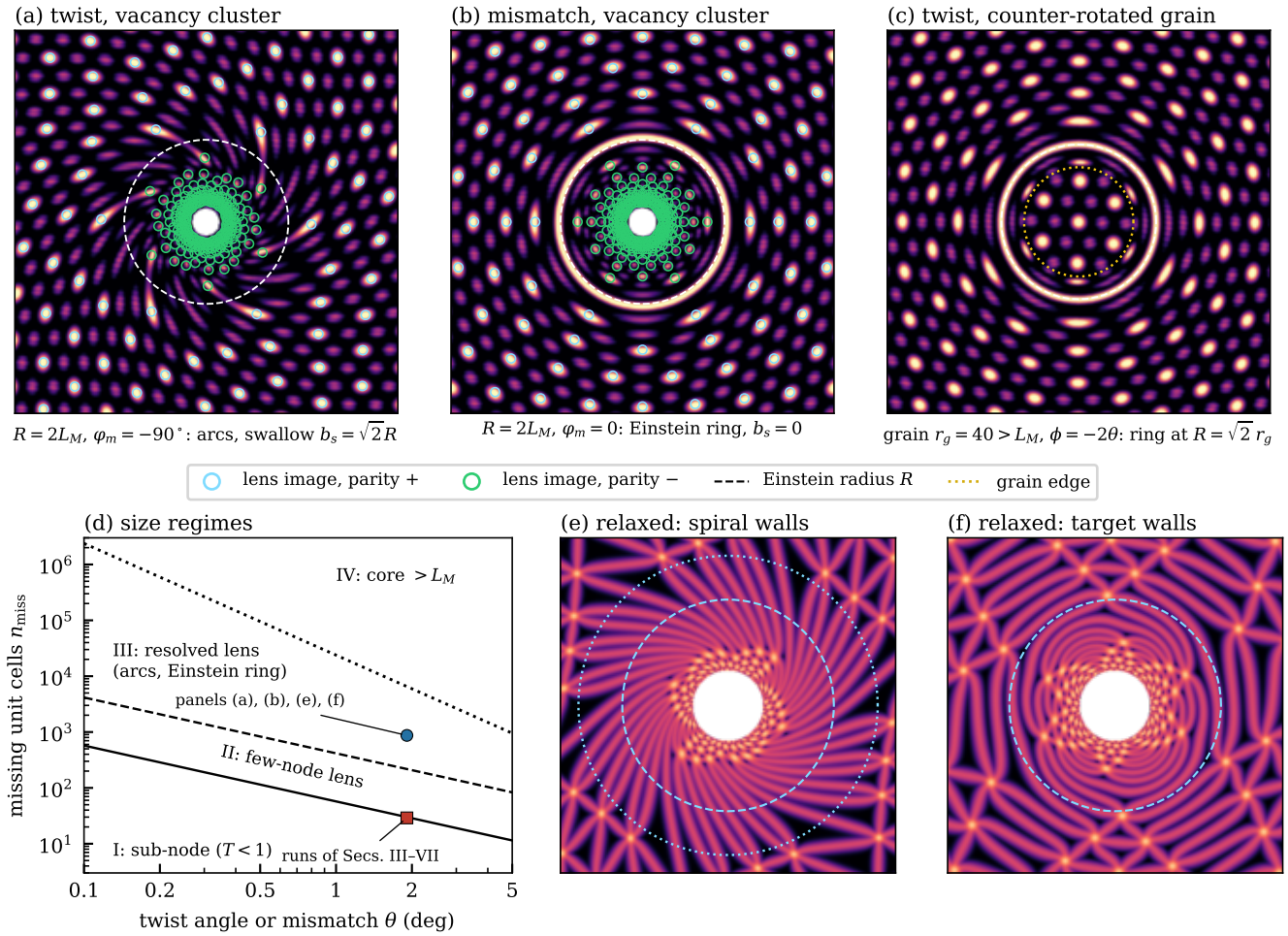


FIG. 12. Lenses larger than the moiré period ($\theta = 1/30$, $L_M = 30$). (a)–(c) Moiré computed from atomic positions as in Fig. 1. (a) Vacancy cluster with $|C| = 120$ ($R = 2L_M$) in a twisted bilayer: images near the critical circle are stretched into arcs, and the region inside $b_s = \sqrt{2}R$ is swallowed. (b) The same defect in a mismatch bilayer ($\varphi_m = 0$): the whole critical circle carries the registry of the defect position (here AA), forming a moiré Einstein ring; inside it the moiré is inverted (green). (c) A grain of radius $r_g = 40 > L_M$ (yellow dotted) rotated by $\phi = -2\theta$ in a twisted bilayer: the grain hosts its own moiré of twist $-\theta$, and an Einstein ring forms at $R = \sqrt{2}r_g$. (d) Size regimes of Eq. (17) in the plane of twist angle and missing unit cells. Square: defects of Figs. 2–11 ($T = 0.97$, at the I/II boundary); circle: panels (a), (b), (e), (f). (e),(f) Relaxed continuum calculation of the defect of (a) and (b) ($r_h = 20$, box half-width 150), shown as $\log_{10} V(u)$ so that domain walls appear as bright lines; dashed: Einstein radius R; dotted in (e): $\sqrt{2}R$. The twisted lens relaxes into a fan of 42 spiral walls around a core binding 37 nodes, which saturates the bound of Eq. (6); the mismatch lens relaxes into nested closed walls (target pattern) inside the Einstein circle.

the auxetic $K/\mu = 0.5$, are model choices. Topological counts and the lens structure do not depend on them, whereas the screening ratio does (Sec. VII). The single-field description neglects layer-resolved displacement and out-of-plane relaxation; these change V and the elastic constants but not the topology or the lens structure. The numerical branch search is not exhaustive, and the nudged-elastic-band barriers [30] we computed for branch switching are upper bounds.

XI. CONCLUSIONS

Moiré nodes are topological charges of the registry map, and atomic area and torque defects act on them as lenses with mass $m = C/\kappa$. On average over its position, a defect missing $n_{\rm miss}$ unit cells removes $\theta n_{\rm miss}$ ordinary nodes; a fraction $1 - \cos\varphi_m$ is swallowed and bound to the core as an integer, and a fraction $\cos\varphi_m$ is pushed to infinity. This explains why area defects are swallowed in twisted bilayers but, depending on the host layer, not in mismatch bilayers, and why closing and opening irises are symmetric in one case and not in the other. The bound integer follows this partition over

the range of lens strengths we tested when the moiré is relaxed from the lens field, but other relaxation histories select different integer branches. Beyond the atomic charge, the far node charge is dressed by the force dipole of the reorganized registry forces. For a single defect this dressing screens the lens: the relaxed moiré overtwists near the core and keeps part of the removed nodes, so the far lens mass is ρm with ρ decreasing from 1 to about 0.1 as relaxation strengthens and as the bulk stiffens, independent of $|C|$. A local-density theory built on the energy of sheared moirés explains this through the high registry energy of the inverted moiré inside the Einstein radius. The moiré crystal has anomalous elastic constants—a negative bulk modulus for twist and a negative rotational stiffness for mismatch—yet the drive of a local source is the microscopic ring traction rather than the moiré pre-stress. When the Einstein radius exceeds the moiré period the lens is resolved into arcs, an inverted inner moiré and, for $\varphi_m = 0$, a ring of uniform stacking; relaxation turns these into spiral or concentric wall textures, enlarges the swallowed region, and dresses the far charge. Open problems include a gradient correction to the local-density theory of screening, the ground state among the coexisting screened states for $\varphi_m \neq 0$, a rule that predicts $C_{\rm reg}$ of interacting defects, placement-averaged responses for sources much larger than L_M, and the node currents pumped by moving defects.

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DATA AVAILABILITY

Code and analysis data are available at https://github.com/cywon-commits/moire-iris-lens.

Appendix A: Equivalence of elastic references

The clamped calculations assign elastic energy to the strain of the total u, i.e., $E_{\rm el}[u] = E_{\rm el}[u_b] + E_{\rm el}[\delta u] + \int \sigma_b : \varepsilon(\delta u)$. For affine u_b the stress σ_b is uniform, and the cross term becomes the boundary integral $\oint \sigma_b n \cdot \delta u$. It vanishes on the outer boundary ($\delta u = 0$). On a core rim σ_b is isotropic (mismatch) or zero (twist), so the cross term couples only to the mean normal displacement, which is fixed by the constraint. The cross term is therefore a constant, and equilibria, energy differences and interactions of iris defects agree between the two references. This equivalence does not extend to the eigen-inclusion of Sec. VIII, whose jump s couples to the background stress through $\int_D \sigma_b : \nabla s$; this term is exactly the placement-averaged mismatch drive $\langle E_1 \rangle$, which vanishes in a stress-free reference.

Appendix B: Removed node charge of a pair

For two defects, $J = 1 - |f'|^2$ with $f' = -m_1/(\bar{z} - \bar{z}_1)^2 - m_2/(\bar{z} - \bar{z}_2)^2$. Averaged over translations, the number of removed ordinary nodes is $n_M \int (1 - J \mathbb{1}_{J>0}) \mathrm{d}^2 z = n_M \int \min(1, |f'|^2) \mathrm{d}^2 z$. For one defect this gives $2\pi R^2 n_M = T$, consistent with Eq. (10). Numerically, $(+,+)$ pairs give 1.79–1.93, and $(+,-)$ pairs give $0.02, 0.93, 1.46, 1.90$ at $d = 0, 8, 16, 30$.

Appendix C: Counting

Nodes: on each grid cell, we count the windings of the polygon formed by the registry values at its four corners around the AA lattice points. Shared edges cancel, so Eq. (5) holds exactly on the grid. $Q_{\rm hole}$*:* winding along a staircase loop just outside the core (radius ≈ 3). z*:* changes of the AB/BA label along circles of radius 5, 7 and 9. *Node positions* for the inclusion analysis are obtained by solving $u(z) = c_{\rm AA} + n$ with Newton iterations on a cubic interpolant (residual $< 10^{-12}$).

Appendix D: Numerical parameters

TABLE VI. Parameters (lattice constant $a = 1$).

twist angle or mismatch θ	1/30
registry V_0, V_3	0.005, $\sqrt{3}V_0$
Lamé μ; λ	1; -0.5 or 1
$K/\mu = (\lambda + \mu)/\mu$	0.5 or 2
$\lvert C\rvert$; core radius r_h; rim penalty	4; 2.4; 200
Einstein radius $R = \sqrt{\lvert C\rvert/\theta}$	≈ 11
grid spacing h	0.45
box half-width	54–200
relaxed moiré period	≈ 31

[1] R. Bistritzer and A. H. MacDonald, Moiré bands in twisted double-layer graphene, Proc. Natl. Acad. Sci. USA **108**, 12233 (2011).

[2] Y. Cao, V. Fatemi, S. Fang, K. Watanabe, T. Taniguchi, E. Kaxiras, and P. Jarillo-Herrero, Unconventional superconductivity in magic-angle graphene superlattices, Nature **556**, 43 (2018).

[3] Y. Cao *et al.*, Correlated insulator behaviour at half-filling in magic-angle graphene superlattices, Nature **556**, 80 (2018).

[4] E. Y. Andrei and A. H. MacDonald, Graphene bilayers with a twist, Nat. Mater. **19**, 1265 (2020).

[5] D. M. Kennes *et al.*, Moiré heterostructures as a condensed-matter quantum simulator, Nat. Phys. **17**, 155 (2021).

[6] A. M. Popov, I. V. Lebedeva, A. A. Knizhnik, Yu. E. Lozovik, and B. V. Potapkin, Commensurate-incommensurate phase transition in bilayer graphene, Phys. Rev. B **84**, 045404 (2011).

[7] J. S. Alden *et al.*, Strain solitons and topological defects in bilayer graphene, Proc. Natl. Acad. Sci. USA **110**, 11256 (2013).

[8] N. N. T. Nam and M. Koshino, Lattice relaxation and energy band modulation in twisted bilayer graphene, Phys. Rev. B **96**, 075311 (2017).

[9] S. Carr, D. Massatt, S. B. Torrisi, P. Cazeaux, M. Luskin, and E. Kaxiras, Relaxation and domain formation in incommensurate two-dimensional heterostructures, Phys. Rev. B **98**, 224102 (2018).

[10] H. Yoo *et al.*, Atomic and electronic reconstruction at the van der Waals interface in twisted bilayer graphene, Nat. Mater. **18**, 448 (2019).

[11] S. Dai, Y. Xiang, and D. J. Srolovitz, Twisted bilayer graphene: Moiré with a twist, Nano Lett. **16**, 5923 (2016).

[12] F. Gargiulo and O. V. Yazyev, Structural and electronic transformation in low-angle twisted bilayer graphene, 2D Mater. **5**, 015019 (2018).

[13] K. Zhang and E. B. Tadmor, Structural and electron diffraction scaling of twisted graphene bilayers, J. Mech. Phys. Solids **112**, 225 (2018).

[14] A. Uri *et al.*, Mapping the twist-angle disorder and Landau levels in magic-angle graphene, Nature **581**, 47 (2020).

[15] A. Kerelsky *et al.*, Maximized electron interactions at the magic angle in twisted bilayer graphene, Nature **572**, 95 (2019).

[16] L. Huder, A. Artaud, T. Le Quang, G. Trambly de Laissardière, A. G. M. Jansen, G. Lapertot, C. Chapelier, and V. T. Renard, Electronic spectrum of twisted graphene layers under heterostrain, Phys. Rev. Lett. **120**, 156405 (2018).

[17] N. P. Kazmierczak *et al.*, Strain fields in twisted bilayer graphene, Nat. Mater. **20**, 956 (2021).

[18] D. Halbertal *et al.*, Moiré metrology of energy landscapes in van der Waals heterostructures, Nat. Commun. **12**, 242 (2021).

[19] C. N. Lau, M. W. Bockrath, K. F. Mak, and F. Zhang, Reproducibility in the fabrication and physics of moiré materials, Nature **602**, 41 (2022).

[20] F. Banhart, J. Kotakoski, and A. V. Krasheninnikov, Structural defects in graphene, ACS Nano **5**, 26 (2011).

[21] A. Einstein, Lens-like action of a star by the deviation of light in the gravitational field, Science **84**, 506 (1936).

[22] P. Schneider, J. Ehlers, and E. E. Falco, *Gravitational Lenses* (Springer, Berlin, 1992).

[23] D. R. Nelson, *Defects and Geometry in Condensed Matter Physics* (Cambridge University Press, Cambridge, 2002).

[24] M. Kléman and J. Friedel, Disclinations, dislocations, and continuous defects: A reappraisal, Rev. Mod. Phys. **80**, 61 (2008).

[25] N. D. Mermin, The topological theory of defects in ordered media, Rev. Mod. Phys. **51**, 591 (1979).

[26] J. D. Eshelby, The determination of the elastic field of an ellipsoidal inclusion, and related problems, Proc. R. Soc. London A **241**, 376 (1957).

[27] L. D. Landau and E. M. Lifshitz, *Theory of Elasticity*, 3rd ed. (Pergamon, Oxford, 1986).

[28] J. Goldstone and F. Wilczek, Fractional quantum numbers on solitons, Phys. Rev. Lett. **47**, 986 (1981).

[29] D. C. Liu and J. Nocedal, On the limited memory BFGS method for large scale optimization, Math. Program. **45**, 503 (1989).

[30] G. Henkelman, B. P. Uberuaga, and H. Jónsson, A climbing image nudged elastic band method for finding saddle points and minimum energy paths, J. Chem. Phys. **113**, 9901 (2000).

[31] B. Butz, C. Dolle, F. Niekiel, K. Weber, D. Waldmann, H. B. Weber, B. Meyer, and E. Spiecker, Dislocations in bilayer graphene, Nature **505**, 533 (2014).

[32] M. Koshino and Y.-W. Son, Moiré phonons in twisted bilayer graphene, Phys. Rev. B **100**, 075416 (2019).

[33] H. Ochoa and R. M. Fernandes, Degradation of phonons in disordered moiré superlattices, Phys. Rev. Lett. **128**, 065901 (2022).

[34] Z. Bi, N. F. Q. Yuan, and L. Fu, Designing flat bands by strain, Phys. Rev. B **100**, 035448 (2019).

[35] K. Chang and S. Refsdal, Flux variations of QSO 0957+561 A, B and image splitting by stars near the light path, Nature **282**, 561 (1979).